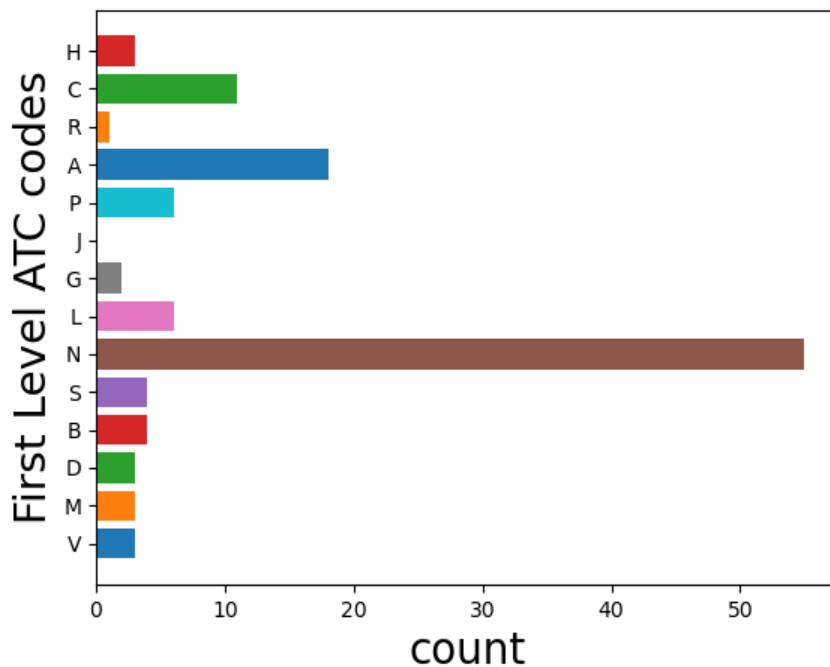


Community/Cluster 1

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Drugs with property N:

{'Phenobarbital', 'Glutethimide', 'Bupropion', 'Ziconotide', 'Nicotine', 'Gabapentin', 'Dalfampridine', 'Methylphenobarbital', 'Amoxapine', 'Felbamate', 'Desflurane', 'Orphenadrine', 'Varenicline', 'Zaleplon', 'Vigabatrin', 'Sevoflurane', 'Butabarbital', 'Primidone', 'Methoxyflurane', 'Isoflurane', 'Halothane', 'Lumateperone', 'Memantine', 'Lamotrigine', 'Dextromethorphan', 'Methyprylon', 'Haloperidol', 'Talbutal', 'Topiramate', 'Butalbital', 'Etomidate', 'Thiopental', 'Valproic acid', 'Methadone', 'Secobarbital', 'Meperidine', 'Ginkgo biloba', 'Amifampridine', 'Enflurane', 'Meprobamate', 'Thiamylal', 'Butobarbital', 'Eszopiclone', 'Zolpidem', 'Phenelzine', 'Levacetylmethadol', 'Pentobarbital', 'Amantadine', 'Esketamine', 'Methohexital', 'Amobarbital', 'Levetiracetam', 'Cannabidiol', 'Disulfiram', 'Zopiclone'}

In [9]:

Drugs not having property N:

['Insulin human', 'Botulinum toxin type B', 'Insulin lispro', 'Insulin glargine', 'Insulin glulisine', 'Insulin pork', 'Pyridoxal phosphate', 'Ascorbic acid', 'Glutamic acid', 'Glycine', 'Sildenafil', 'Isradipine', 'Pimecrolimus', 'Amlodipine', 'Carisoprodol', 'Lindane', 'Atropine', 'Verapamil', 'Loperamide', 'Levamisole', 'Sirolimus', 'Ethanol', 'Azathioprine', 'Felodipine', 'Nitrendipine', 'Flumazenil', 'Bepidil', 'Mecasermin', 'Insulin detemir', 'Insulin aspart', 'Everolimus', 'Zinc', 'Propanoic acid', 'Eflornithine', 'Temsilolimus', 'Teprotumumab', 'Insulin beef', 'Somatrem', 'Ubidecarenone', 'Zinc sulfate', 'Insulin degludec', 'Calcium Phosphate', 'Artenimol', 'Fostamatinib', 'Brigatinib', 'Fluciclovine (18F)', 'Bioallethrin', 'Zinc acetate', 'Zinc chloride', 'Mecasermin rinfabate', 'Insulin human', 'Botulinum toxin type B', 'Insulin lispro', 'Insulin glargine', 'Insulin pork', 'Pyridoxal phosphate', 'Ascorbic acid', 'Glutamic acid', 'Glycine', 'Sildenafil', 'Isradipine', 'Pimecrolimus', 'Carisoprodol', 'Amlodipine', 'Lindane', 'Atropine', 'Verapamil', 'Loperamide', 'Levamisole', 'Sirolimus', 'Ethanol', 'Azathioprine', 'Felodipine', 'Nitrendipine', 'Flumazenil', 'Bepidil', 'Mecasermin', 'Insulin aspart', 'Insulin detemir', 'Insulin glulisine', 'Everolimus', 'Zinc', 'Propanoic acid', 'Eflornithine', 'Temsilolimus', 'Teprotumumab', 'Somatrem', 'Ubidecarenone', 'Zinc sulfate', 'Insulin beef', 'Insulin degludec', 'Calcium Phosphate', 'Artenimol', 'Fostamatinib', 'Brigatinib', 'Fluciclovine (18F)', 'Bioallethrin', 'Zinc acetate', 'Zinc chloride', 'Mecasermin rinfabate']

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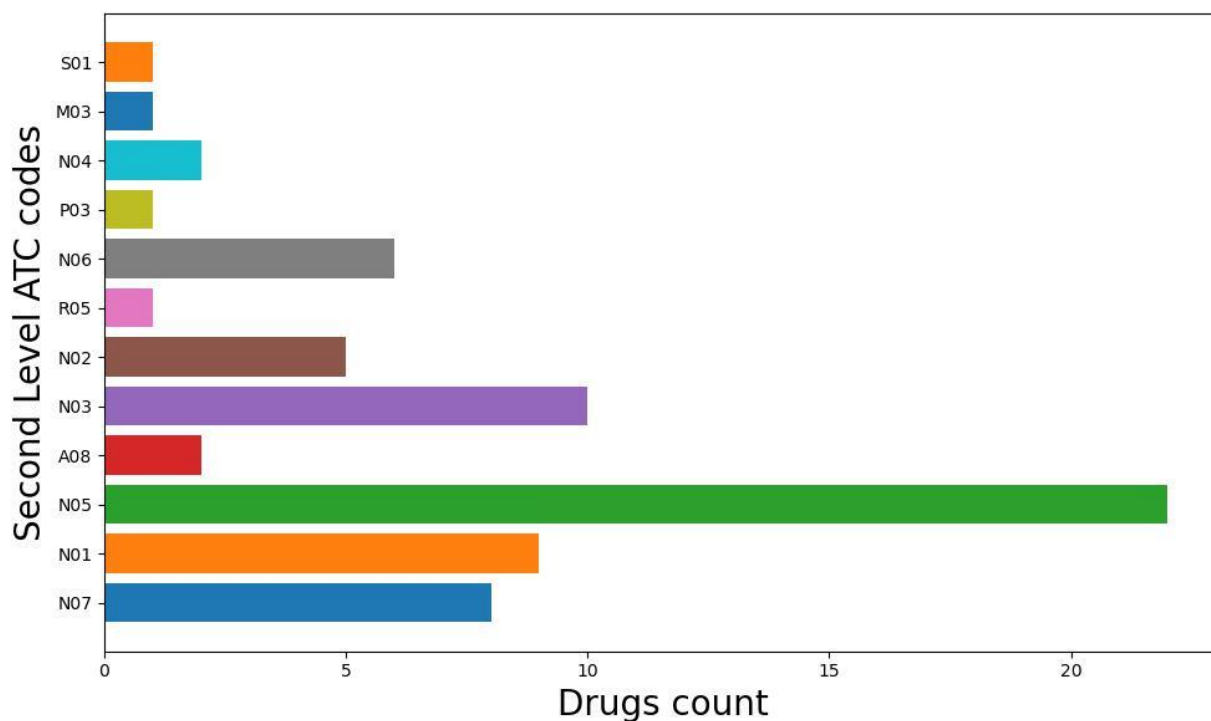
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Level 2 on N:

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Drug Repositioning Pipeline Integrating Community Analysis in Drug-Drug Similarity Networks and Automated ATC Community Labeling to Enhance Molecular Docking Validation

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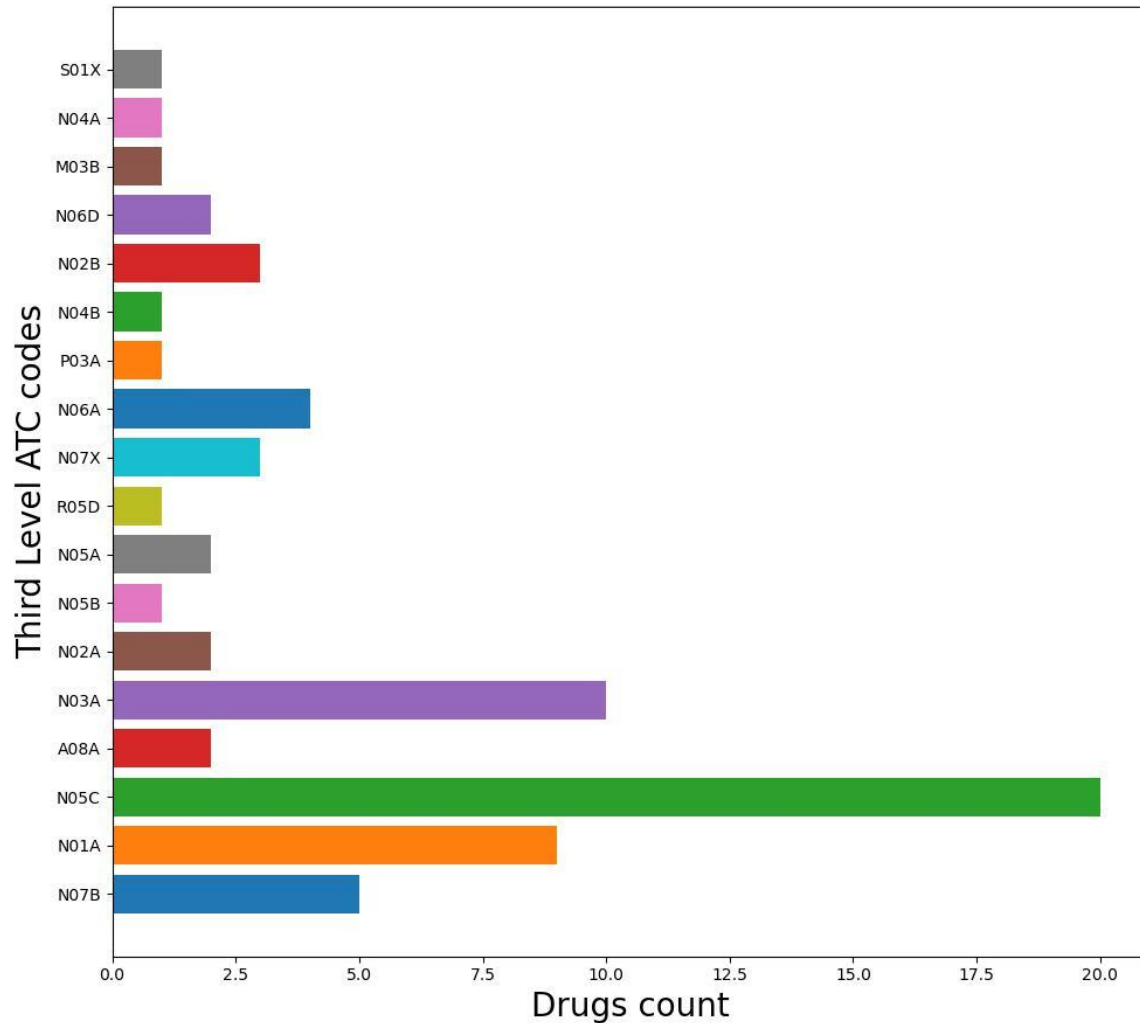


Level 3 on N:

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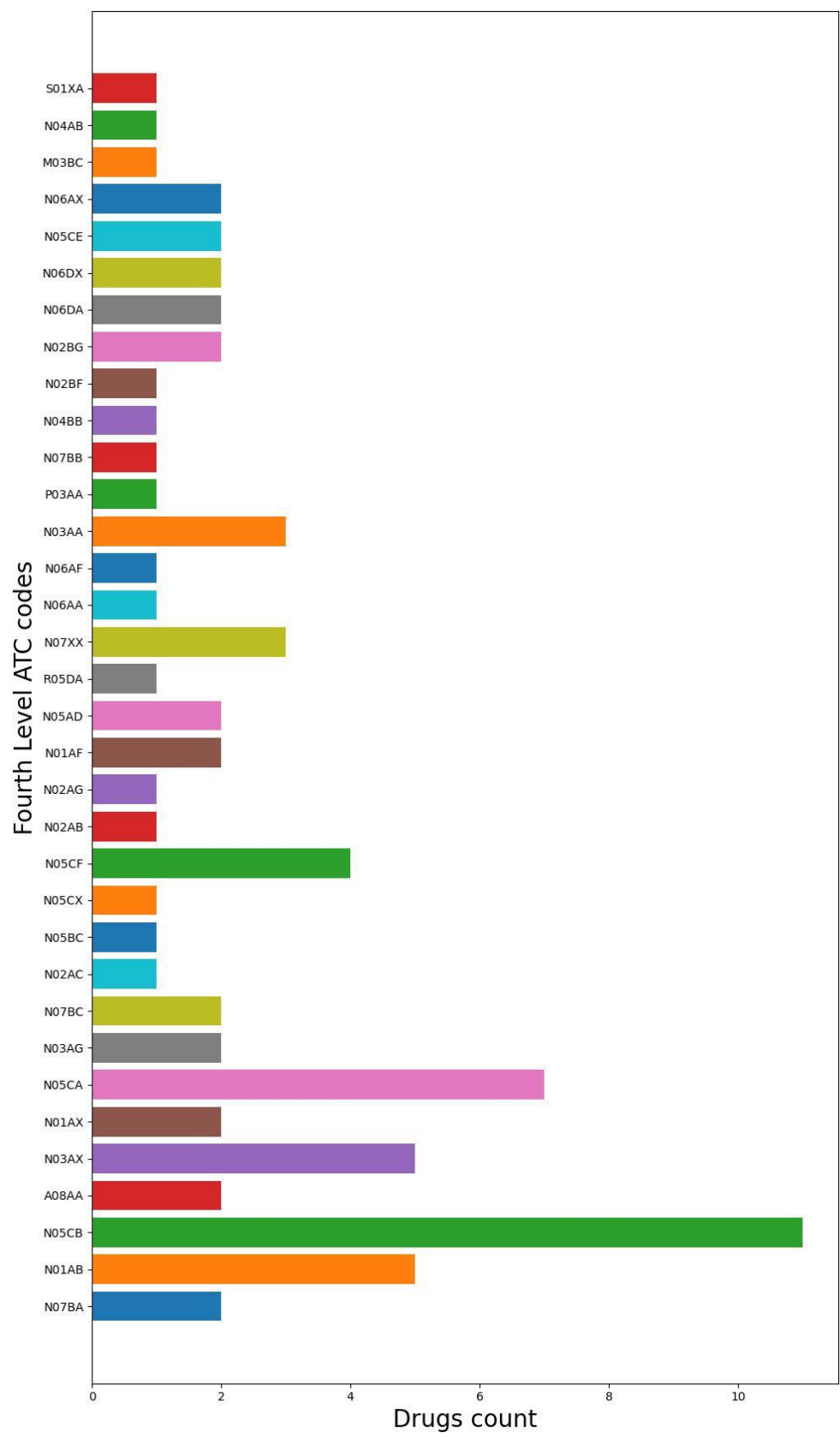


Level 4 on N:

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Drug Repositioning Pipeline Integrating Community Analysis in Drug-Drug Similarity Networks and Automated ATC Community Labeling to Enhance Molecular Docking Validation

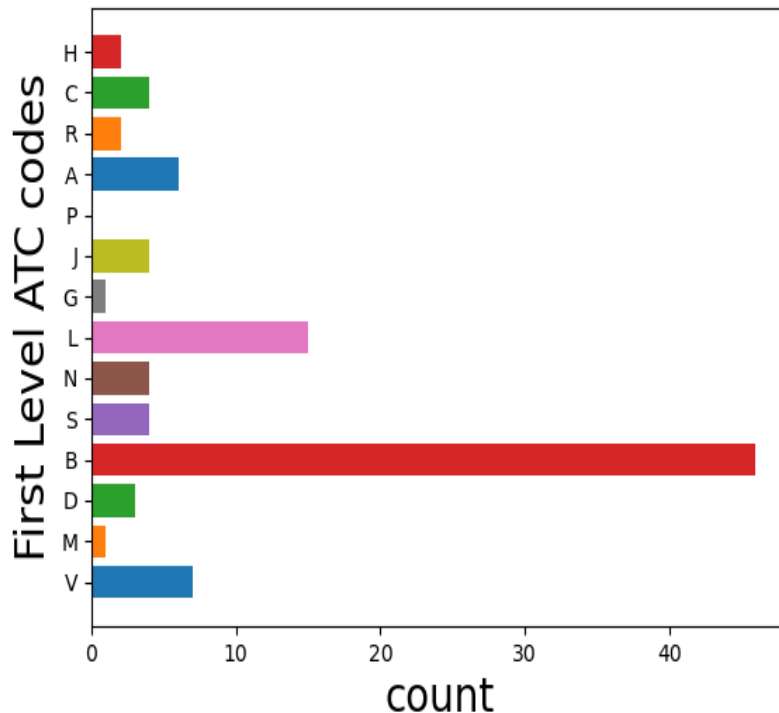
D. Colibășanu, V. Groza, M.A. Occhiuzzi, F. Grande, M. Udrescu, L. Udrescu



Community/Cluster 2

{'V': {'Florbetapir (18F)', 'Dimercaprol', 'Flutemetamol (18F)', 'Glutathione', 'Florbetaben (18F)', 'Deferoxamine', 'Benzoic acid'}, 'M': {'Collagenase clostridium histolyticum'}, 'D': {'Collagenase clostridium histolyticum', 'Tetracycline', 'Minocycline'}, 'B': {'Von Willebrand factor human', 'Anti-inhibitor coagulant complex', 'Antithrombin III human', 'Lanadelumab', 'Menadione', 'Catridecacog', 'Ecallantide', 'Drotrecogin alfa', 'Human C1-esterase inhibitor', 'Urokinase', 'Antihemophilic factor human', 'Tromethamine', 'Fondaparinux', 'Streptokinase', 'Coagulation factor VIIa Recombinant Human', 'Apixaban', 'Berotralstat', 'Argatroban', 'Tenecteplase', 'Aminocaproic acid', 'Bemiparin', 'Sulodexide', 'Conestat alfa', 'Dabigatran etexilate', 'Alteplase', 'Coagulation Factor IX Human', 'Reteplase', 'Aprotinin', 'Edoxaban', 'Rivaroxaban', 'Betrixaban', 'Tinzaparin', 'Enoxaparin', 'Bivalirudin', 'Ximelagatran', 'Tranexamic acid', 'Danaparoid', 'Protein C', 'Dalteparin', 'Lepirudin', 'Thrombin', 'Ancrod', 'Phylloquinone', 'Emicizumab', 'Coagulation factor VII human', 'Anistreplase'}, 'S': {'Bevacizumab', 'Tetracycline', 'Ocriplasmin', 'Alteplase'}, 'N': {'Aducanumab', 'Tafamidis', 'Ezogabine', 'Stiripentol'}, 'L': {'Cetuximab', 'Eculizumab', 'Sarilumab', 'Etanercept', 'Alefcept', 'Bevacizumab', 'Pegcetacoplan', 'Diethylstilbestrol', 'Vorinostat', 'Porfimer sodium', 'Gemtuzumab ozogamicin', 'Sutimlimab', 'Pralsetinib', 'Alemtuzumab', 'Ravulizumab'}, 'G': {'Diethylstilbestrol'}, 'J': {'Minocycline', 'Tetracycline', 'Ribostamycin', 'Palivizumab'}, 'P': set(), 'A': {'Minocycline', 'Nicotinamide', 'Aluminium phosphate', 'Tetracycline', 'Benzoic acid', 'Migalastat'}, 'R': {'Benralizumab', 'Isoprenaline'}, 'C': {'Ethanolamine oleate', 'Volanesorsen', 'Isoprenaline', 'Dopamine'}, 'H': {'Etelcalcetide', 'Cinacalcet'}}

In [24]:



Drugs with property B:

['Ancrod', 'Anistreplase', 'Phylloquinone', 'Ecallantide', 'Sulodexide', 'Human C1-esterase inhibitor', 'Antithrombin III human', 'Von Willebrand factor human', 'Ximelagatran', 'Danaparoid', 'Lanadelumab', 'Tranexamic acid', 'Bemiparin', 'Reteplase', 'Aprotinin', 'Thrombin', 'Catridecacog', 'Menadione', 'Apixaban', 'Dalteparin', 'Protein C', 'Argatroban', 'Betrixaban', 'Berotralstat', 'Rivaroxaban', 'Tinzaparin', 'Coagulation Factor IX Human', 'Coagulation factor VIIa Recombinant Human', 'Conestat alfa', 'Emicizumab', 'Edoxaban', 'Alteplase', 'Enoxaparin', 'Tenecteplase', 'Dabigatran etexilate', 'Bivalirudin', 'Anti-inhibitor coagulant complex', 'Antihemophilic factor human', 'Coagulation factor VII human', 'Aminocaproic acid', 'Tromethamine', 'Drotrecogin alfa', 'Streptokinase', 'Fondaparinux', 'Lepirudin', 'Urokinase']

Drugs not having property B:

['Cetuximab', 'Etanercept', 'Collagenase clostridium histolyticum', 'Gemtuzumab ozogamicin', 'Alemtuzumab', 'Alefcept', 'Palivizumab', 'Bevacizumab', 'Glutathione', 'Diethylstilbestrol', 'Porfimer sodium', 'Deferoxamine', 'Tetracycline', 'Dopamine', 'Cinacalcet', 'Minocycline', 'Isoprenaline', 'Eculizumab', 'Vorinostat', 'Nicotinamide', 'Ribostamycin', 'Benzoic acid', 'Ezogabine', 'Migalastat', 'Ethanolamine oleate', 'Dimercaprol', 'Ocriplasmin', 'Stiripentol', 'Florbetaben (18F)', 'Florbetapir (18F)', 'Flutemetamol (18F)', 'Ravulizumab', 'Tafamidis', 'Sarilumab', 'Benralizumab', 'Aducanumab', 'Etelcalcetide', 'Aluminium phosphate', 'Sutimlimab', 'Volanesorsen', 'Pralsetinib', 'Pegcetacoplan', 'Cetuximab', 'Etanercept', 'Collagenase clostridium histolyticum', 'Gemtuzumab ozogamicin', 'Alemtuzumab', 'Alefcept', 'Palivizumab', 'Bevacizumab', 'Glutathione', 'Diethylstilbestrol', 'Ocriplasmin', 'Porfimer sodium', 'Deferoxamine', 'Tetracycline', 'Dopamine', 'Cinacalcet', 'Minocycline', 'Isoprenaline', 'Eculizumab', 'Vorinostat', 'Nicotinamide', 'Ribostamycin', 'Benzoic acid', 'Migalastat', 'Ezogabine', 'Ethanolamine oleate', 'Dimercaprol', 'Stiripentol', 'Florbetaben (18F)', 'Florbetapir (18F)', 'Flutemetamol (18F)', 'Ravulizumab', 'Tafamidis', 'Sarilumab', 'Aducanumab', 'Benralizumab', 'Etelcalcetide', 'Aluminium phosphate', 'Volanesorsen', 'Sutimlimab', 'Pralsetinib', 'Pegcetacoplan']

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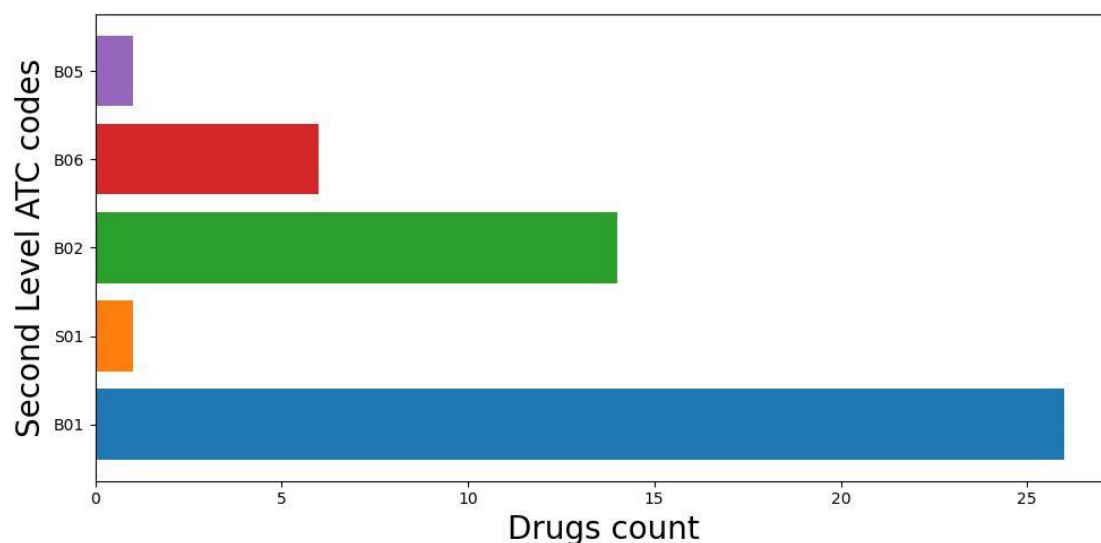
In [7]:

Level 2 on B:

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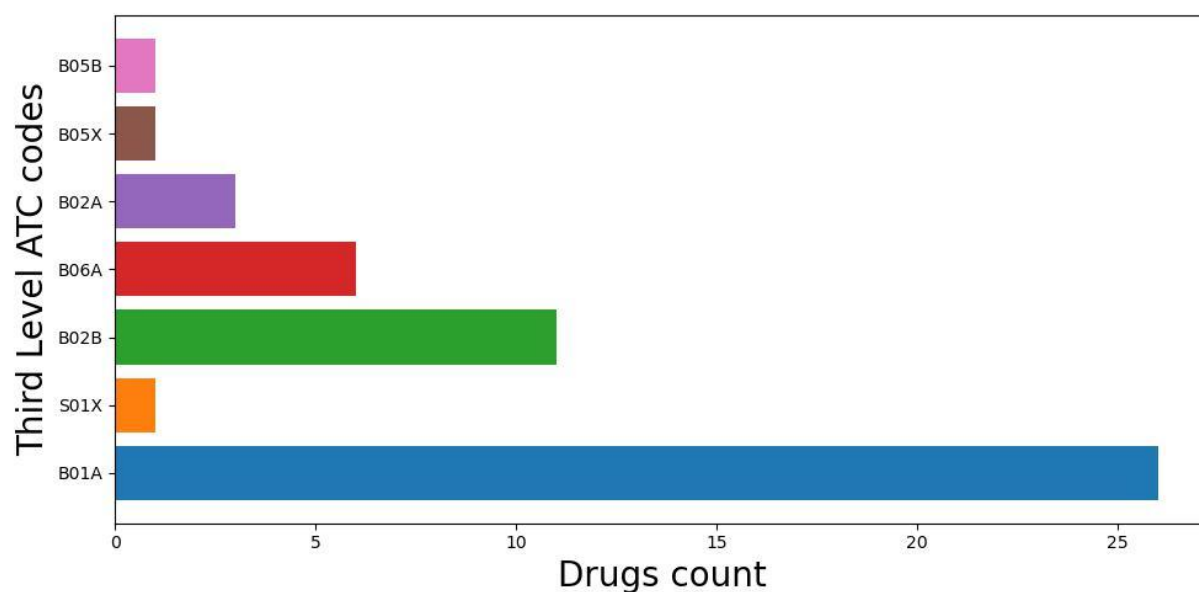
Drug Repositioning Pipeline Integrating Community Analysis in Drug-Drug Similarity Networks and Automated ATC Community Labeling to Enhance Molecular Docking Validation

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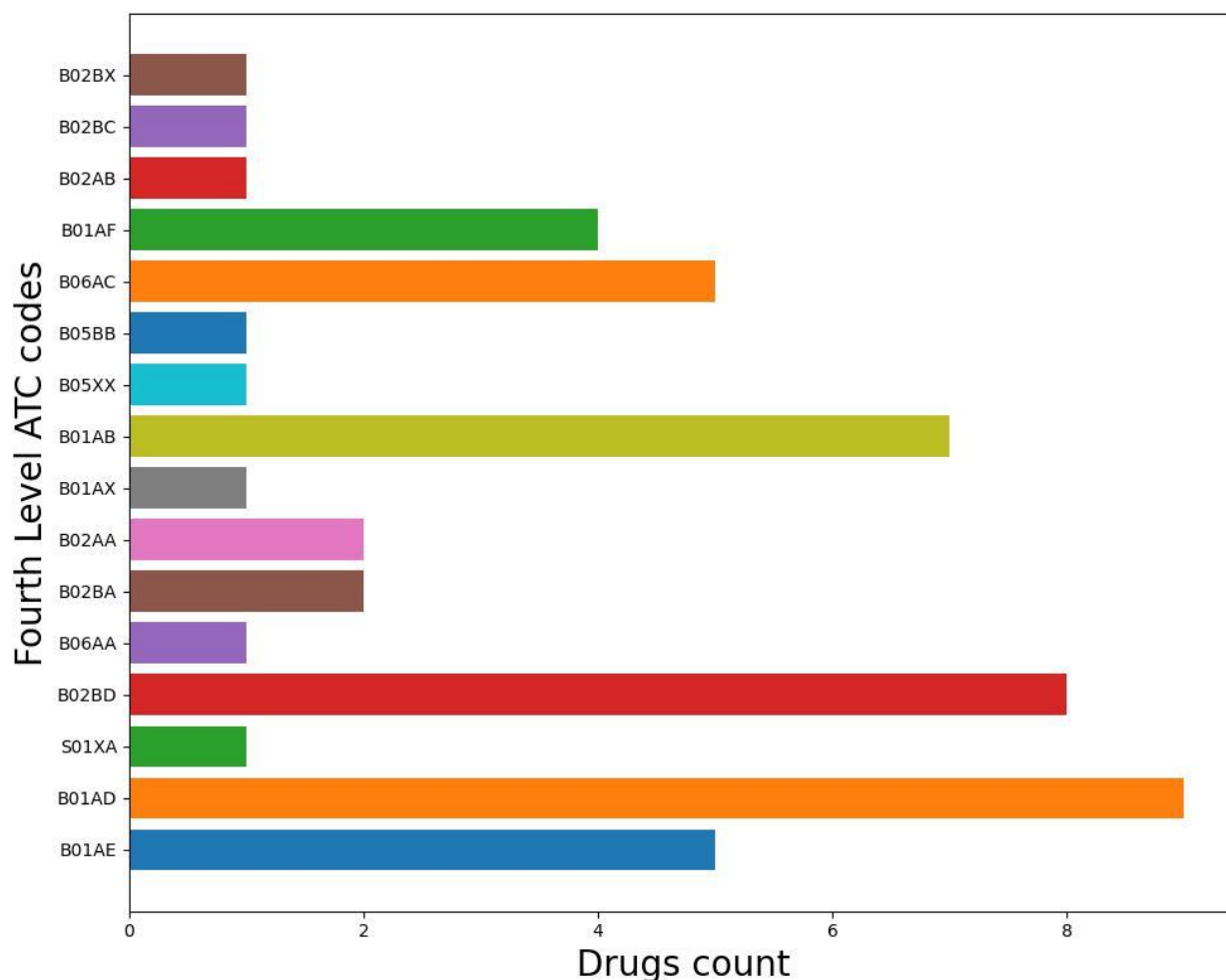
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Level 4 on B:

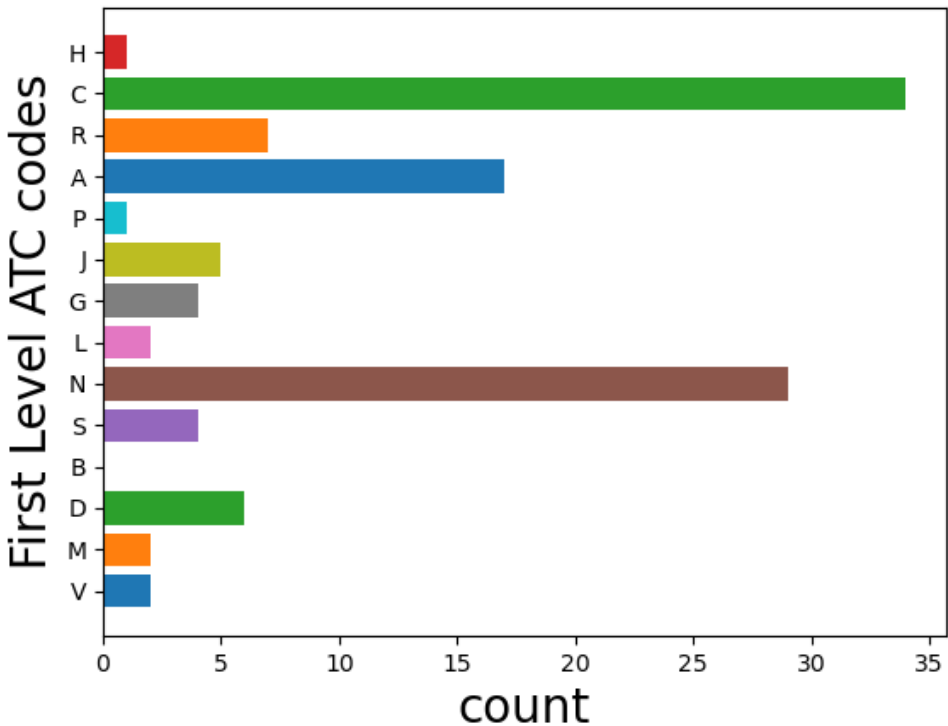
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Community/Cluster 3

```
{'V': {'Tolbutamide', 'Promethazine'}, 'M': {'Clodronic acid', 'Etidronic acid'}, 'D': {'Cinchocaine', 'Doxepin', 'Promethazine', 'Ketoconazole', 'Lidocaine', 'Erythromycin'}, 'B': set(), 'S': {'Erythromycin', 'Cinchocaine', 'Ciprofloxacin', 'Lidocaine'}, 'N': {'Fluphenazine', 'Prilocaine', 'Trifluoperazine', 'Ethotoin', 'Methsuximide', 'Phenytoin', 'Fluoxetine', 'Ethosuximide', 'Thioridazine', 'Mephenytoin', 'Fluvoxamine', 'Melatonin', 'Perphenazine', 'Cinchocaine', 'Desipramine', 'Doxepin', 'Imipramine', 'Pimozide', 'Lidocaine', 'Chlorpromazine', 'Trimethadione', 'Flunarizine', 'Nefazodone', 'Cinnarizine', 'Pitolisant', 'Hydroxyzine', 'Sertindole', 'Fosphenytoin', 'Riluzole'}, 'L': {'Amsacrine', 'Tamoxifen'}, 'G': {'Glymidine', 'Propiverine', 'Ketoconazole', 'Terazosin'}, 'J': {'Ciprofloxacin', 'Isavuconazole', 'Clarithromycin', 'Ketoconazole', 'Erythromycin'}, 'P': {'Halofantrine'}, 'A': {'Chlorobutanol', 'Tolbutamide', 'Gliquidone', 'Phenformin', 'Gliclazide', 'Calcium levulinate', 'Repaglinide', 'Glymidine', 'Nateglinide', 'Calcium citrate', 'Glipizide', 'Acetohexamide', 'Clarithromycin', 'Chlorpropamide', 'Cisapride', 'Levocarnitine', 'Glimepiride'}, 'R': {'Loratadine', 'Benzonatate', 'Terfenadine', 'Pentoxyverine', 'Promethazine', 'Lidocaine', 'Astemizole'}, 'C': {'Procainamide', 'Carvedilol', 'Encainide', 'Nimodipine', 'Quinidine', 'Diltiazem', 'Flecainide', 'Levamlodipine', 'Cinchocaine', 'Nicardipine', 'Ajmaline', 'Clevindipine', 'Disopyramide', 'Bethanidine', 'Lidocaine', 'Nisoldipine', 'Prazosin', 'Vernakalant', 'Levosimendan', 'Phenoxybenzamine', 'Nilvadipine', 'Prenylamine', 'Moricizine', 'Manidipine', 'Sotalol', 'Nifedipine', 'Propafenone', 'Nicorandil', 'Mexiletine', 'Perhexiline', 'Ibutilide', 'Dronedarone', 'Tocainide', 'Dofetilide'}, 'H': {'Ketoconazole'}}
```

In [3]:



Drugs with property C:

['Procainamide', 'Carvedilol', 'Encainide', 'Nimodipine', 'Quinidine', 'Diltiazem', 'Flecainide', 'Levamlodipine', 'Cinchocaine', 'Nicardipine', 'Ajmaline', 'Clevidipine', 'Disopyramide', 'Bethanidine', 'Lidocaine', 'Nisoldipine', 'Prazosin', 'Vernakalant', 'Levosimendan', 'Phenoxybenzamine', 'Nilvadipine', 'Prenylamine', 'Moricizine', 'Manidipine', 'Sotalol', 'Nifedipine', 'Propafenone', 'Nicorandil', 'Mexiletine', 'Perhexiline', 'Ibutilide', 'Dronedarone', 'Tocainide', 'Dofetilide']

In [5]:

Drugs not having property C:

['Fluvoxamine', 'Erythromycin', 'Glimepiride', 'Phenytoin', 'Amsacrine', 'Terfenadine', 'Trimethadione', 'Acetohexamide', 'Loratadine', 'Imipramine', 'Fluoxetine', 'Chlorpromazine', 'Mephenytoin', 'Ciprofloxacin', 'Hydroxyzine', 'Cinnarizine', 'Levocarnitine', 'Ethosuximide', 'Cisapride', 'Fluphenazine', 'Astemizole', 'Chlorpropamide', 'Tamoxifen', 'Thioridazine', 'Clodronic acid', 'Nateglinide', 'Riluzole', 'Prilocaine', 'Ethotoin', 'Trifluoperazine', 'Perphenazine', 'Benzonatate', 'Repaglinide', 'Phenformin', 'Ketoconazole', 'Melatonin', 'Glipizide', 'Promethazine', 'Etidronic acid', 'Pimozide', 'Gliclazide', 'Tolbutamide', 'Doxepin', 'Nefazodone', 'Desipramine', 'Terazosin', 'Clarithromycin', 'Halofantrine', 'Gliquidone', 'Fosphenytoin', 'Glymidine', 'Flunarizine', 'Methsuximide', 'Sertindole', 'Calcium citrate', 'Pentoxyverine', 'Chlorobutanol', 'Isavuconazole', 'Pitolisant', 'Propiverine', 'Calcium levulinate', 'Acetohexamide', 'Fluvoxamine', 'Erythromycin', 'Glimepiride', 'Phenytoin', 'Amsacrine', 'Terfenadine', 'Trimethadione', 'Loratadine', 'Imipramine', 'Fluoxetine', 'Chlorpromazine', 'Mephenytoin', 'Ciprofloxacin', 'Hydroxyzine', 'Cinnarizine', 'Levocarnitine', 'Ethosuximide', 'Chlorpropamide', 'Cisapride', 'Fluphenazine', 'Astemizole', 'Phenformin', 'Tamoxifen', 'Thioridazine', 'Clodronic acid', 'Nateglinide', 'Riluzole', 'Prilocaine', 'Ethotoin', 'Trifluoperazine', 'Perphenazine', 'Benzonatate', 'Repaglinide', 'Ketoconazole', 'Melatonin', 'Glipizide', 'Promethazine', 'Etidronic acid', 'Pimozide', 'Gliclazide', 'Tolbutamide', 'Doxepin', 'Nefazodone', 'Desipramine', 'Terazosin', 'Clarithromycin', 'Halofantrine', 'Gliquidone', 'Fosphenytoin', 'Glymidine', 'Flunarizine', 'Methsuximide', 'Sertindole', 'Calcium citrate', 'Pentoxyverine', 'Chlorobutanol', 'Isavuconazole', 'Pitolisant', 'Propiverine', 'Calcium levulinate']

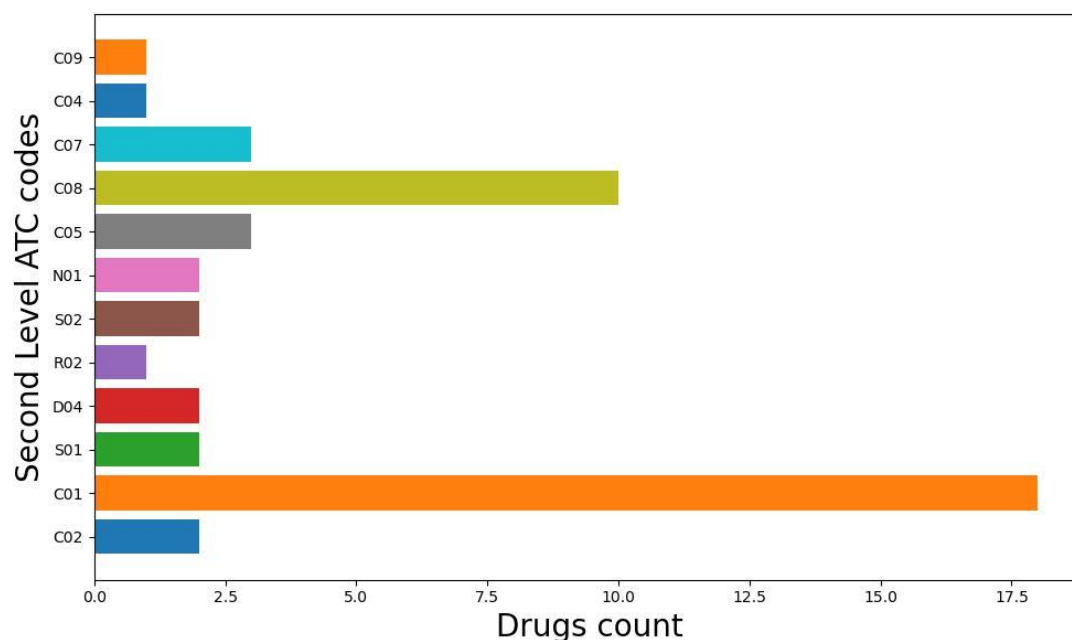
Accuracy: 35.78947368421053

Level 2 on C:

{'C01': {'Nicorandil', 'Ajmaline', 'Moricizine', 'Procainamide', 'Prenylamine', 'Dronedarone', 'Encainide', 'Lidocaine', 'Disopyramide', 'Ibutilide', 'Propafenone', 'Quinidine', 'Vernakalant', 'Mexiletine', 'Tocainide', 'Flecainide', 'Levosimendan', 'Dofetilide'}, 'C02': {'Prazosin', 'Bethanidine'}, 'S01': {'Cinchocaine', 'Lidocaine'}, 'D04': {'Cinchocaine', 'Lidocaine'}, 'R02': {'Lidocaine'}, 'S02': {'Cinchocaine', 'Lidocaine'}, 'N01': {'Cinchocaine', 'Lidocaine'}, 'C05': {'Cinchocaine', 'Diltiazem', 'Lidocaine'}, 'C08': {'Nilvadipine', 'Nicardipine', 'Clevidipine', 'Perhexiline', 'Manidipine', 'Nifedipine', 'Nisoldipine', 'Nimodipine', 'Diltiazem', 'Levamlodipine'}, 'C07': {'Nifedipine', 'Sotalol', 'Carvedilol'}, 'C04': {'Phenoxybenzamine'}, 'C09': {'Manidipine'}}

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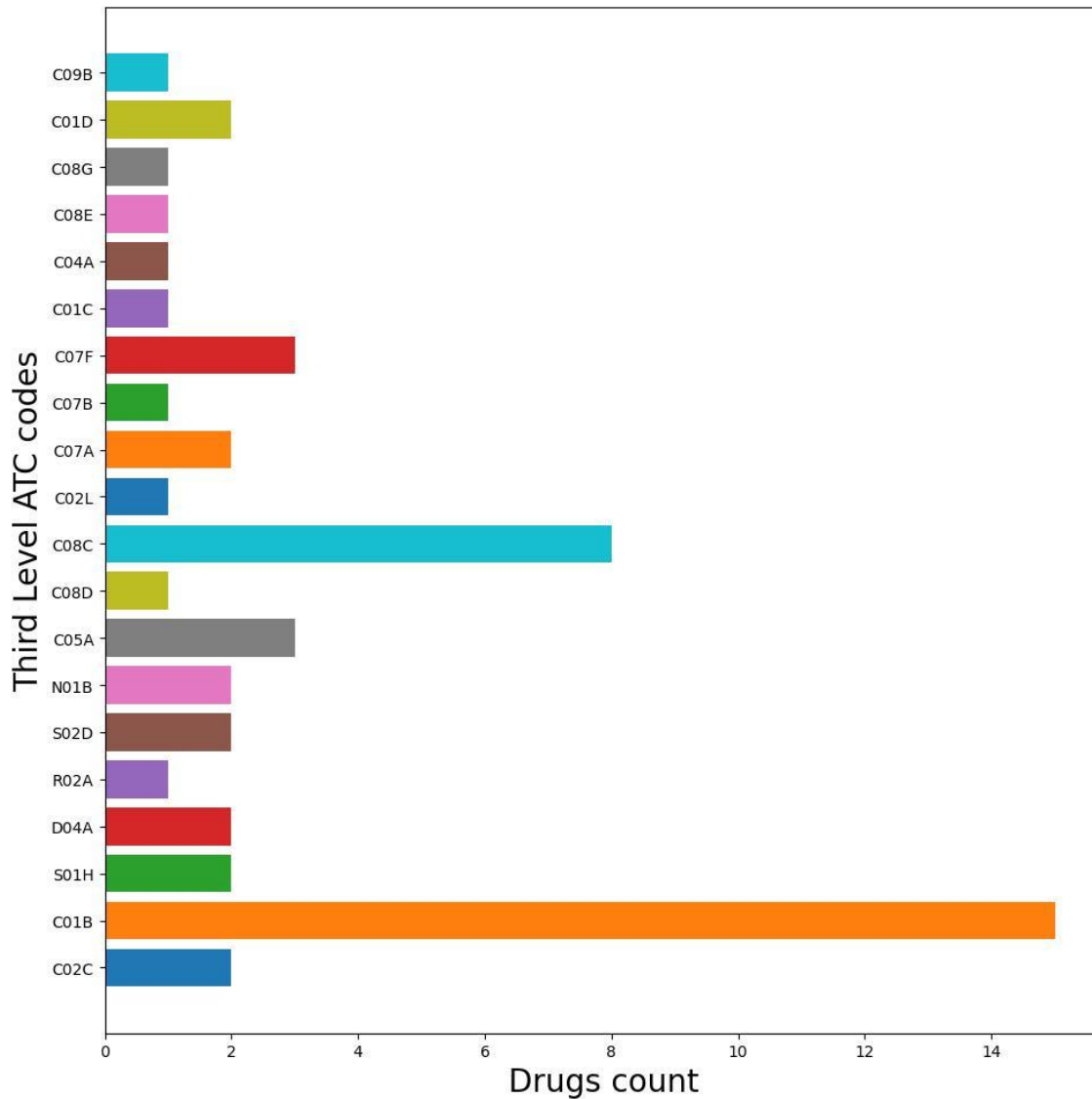


Level 3 on C:

{'C01B': {'Ajmaline', 'Morizine', 'Procainamide', 'Dronedaron', 'Encainide', 'Lidocaine', 'Disopyramide', 'Ibutilide', 'Propafenone', 'Quinidine', 'Vernakalant', 'Mexiletine', 'Tocainide', 'Flecainide', 'Dofetilide'}, 'C02C': {'Prazosin', 'Bethanidine'}, 'S01H': {'Cinchocaine', 'Lidocaine'}, 'D04A': {'Cinchocaine', 'Lidocaine'}, 'R02A': {'Lidocaine'}, 'S02D': {'Cinchocaine', 'Lidocaine'}, 'N01B': {'Cinchocaine', 'Lidocaine'}, 'C05A': {'Cinchocaine', 'Diltiazem', 'Lidocaine'}, 'C08D': {'Diltiazem'}, 'C08C': {'Nilvadipine', 'Nicardipine', 'Clevindipine', 'Manidipine', 'Nifedipine', 'Nisoldipine', 'Nimodipine', 'Levamlodipine'}, 'C02L': {'Prazosin'}, 'C07A': {'Sotalol', 'Carvedilol'}, 'C07B': {'Sotalol'}, 'C07F': {'Nifedipine', 'Sotalol', 'Carvedilol'}, 'C01C': {'Levosimendan'}, 'C04A': {'Phenoxybenzamine'}, 'C08E': {'Perhexiline'}, 'C08G': {'Nifedipine'}, 'C01D': {'Nicorandil', 'Prenylamine'}, 'C09B': {'Manidipine'}}

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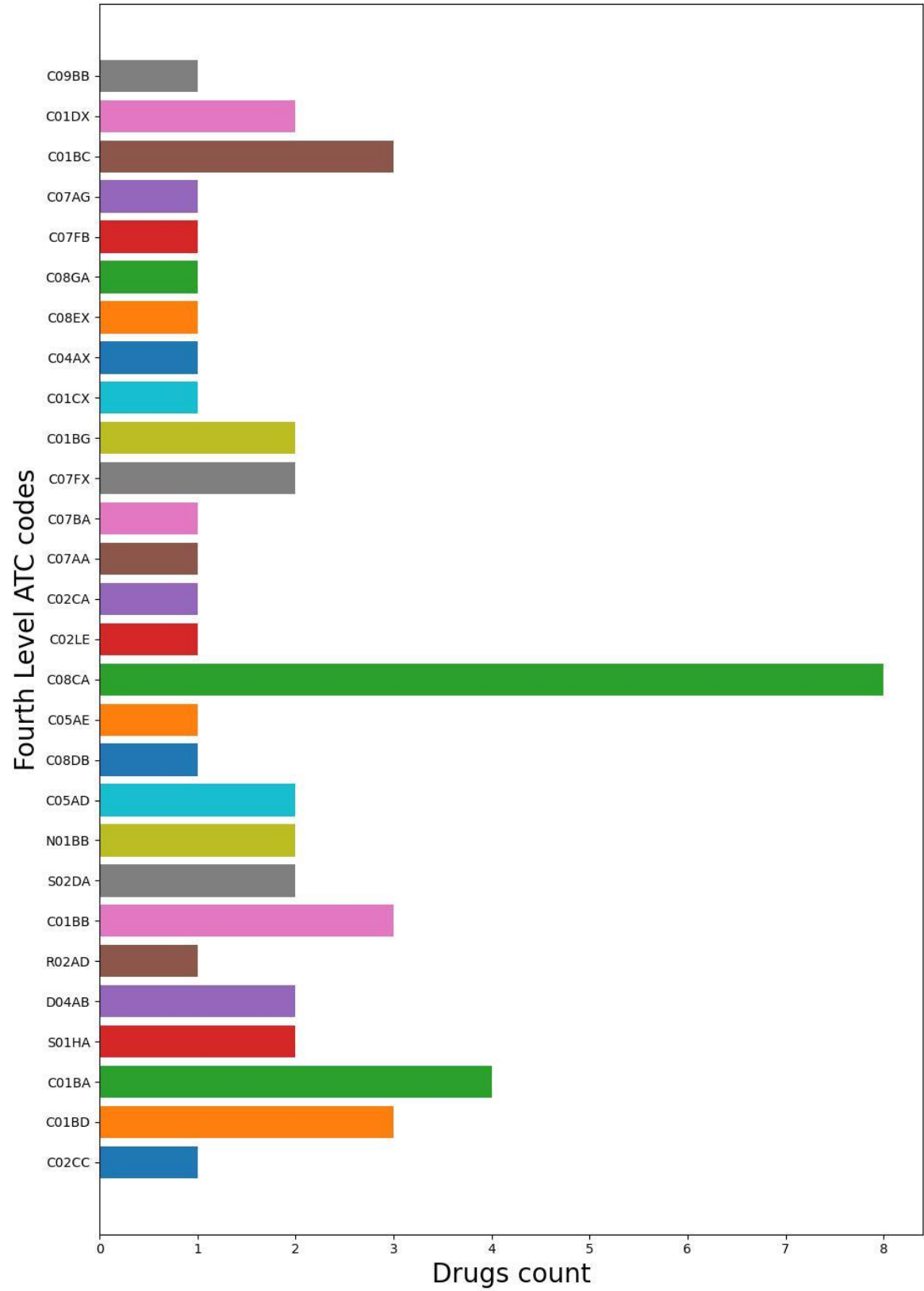


Level 4 on C:

{'C01BD': {'Ibutilide', 'Dronedarone', 'Dofetilide'}, 'C02CC': {'Bethanidine'}, 'C01BA': {'Quinidine', 'Procainamide', 'Disopyramide', 'Ajmaline'}, 'S01HA': {'Cinchocaine', 'Lidocaine'}, 'D04AB': {'Cinchocaine', 'Lidocaine'}, 'R02AD': {'Lidocaine'}, 'C01BB': {'Mexiletine', 'Tocainide', 'Lidocaine'}, 'S02DA': {'Cinchocaine', 'Lidocaine'}, 'N01BB': {'Cinchocaine', 'Lidocaine'}, 'C05AD': {'Cinchocaine', 'Lidocaine'}, 'C08DB': {'Diltiazem'}, 'C05AE': {'Diltiazem'}, 'C08CA': {'Nilvadipine', 'Nicardipine', 'Clevindipine', 'Manidipine', 'Nifedipine', 'Nisoldipine', 'Nimodipine', 'Levamlodipine'}, 'C02LE': {'Prazosin'}, 'C02CA': {'Prazosin'}, 'C07AA': {'Sotalol'}, 'C07BA': {'Sotalol'}, 'C07FX': {'Sotalol', 'Carvedilol'}, 'C01BG': {'Morcizine', 'Vernakalant'}, 'C01CX': {'Levosimendan'}, 'C04AX': {'Phenoxybenzamine'}, 'C08EX': {'Perhexiline'}, 'C08GA': {'Nifedipine'}, 'C07FB': {'Nifedipine'}, 'C07AG': {'Carvedilol'}, 'C01BC': {'Flecainide', 'Propafenone', 'Encainide'}, 'C01DX': {'Nicorandil', 'Prenylamine'}, 'C09BB': {'Manidipine'}}

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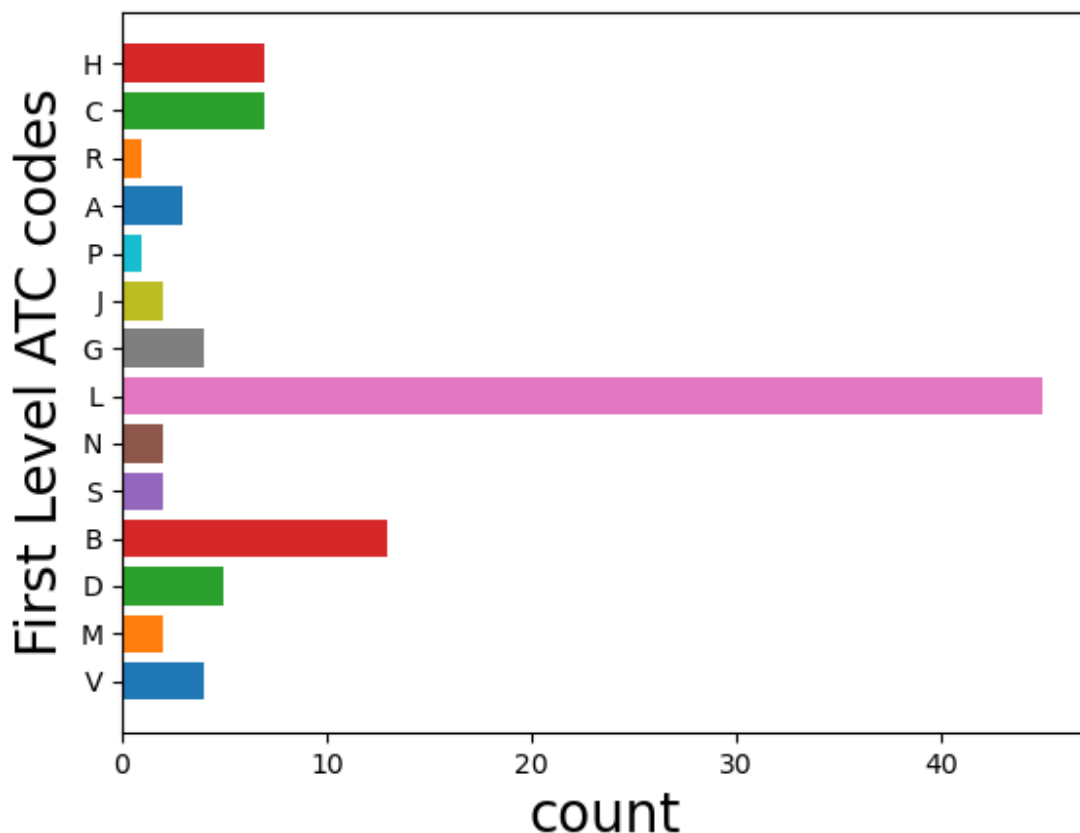
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Community/Cluster 4:

{'V': {'Gonadorelin', 'Nitrous acid', 'Cholecystokinin', 'Palifermin'}, 'M': {'Celecoxib', 'Diboterminalfa'}, 'D': {'Hexachlorophene', 'Halcinonide', 'Becaplermin', 'Aluminum chloride', 'Ingenol mebutate'}, 'B': {'Ferrous ascorbate', 'Ferrous succinate', 'Ferrous glycine sulfate', 'Heparin', 'Darbepoetin alfa', 'Methoxy polyethylene glycol-epoetin beta', 'Iron', 'Ferrous gluconate', 'Voxelotor', 'Peginesatide', 'Ferrous fumarate', 'Erythropoietin', 'Roxadustat'}, 'S': {'Heparin', 'Ampicillin'}, 'N': {'Celecoxib', 'Lacosamide'}, 'L': {'Avapritinib', 'Regorafenib', 'Asciminib', 'Nilotinib', 'Sotorasib', 'Lenvatinib', 'Goserelin', 'Celecoxib', 'Triptorelin', 'Leuprolide', 'Degarelix', 'Pemigatinib', 'Glasdegib', 'Midostaurin', 'Cobimetinib', 'Encorafenib', 'Selpercatinib', 'Infigratinib', 'Leflunomide', 'Erdafitinib', 'Teriflunomide', 'Buserelin', 'Cisplatin', 'Trametinib', 'Vismodegib', 'Sonidegib', 'Tivozanib', 'Dabrafenib', 'Binimetinib', 'Pexidartinib', 'Futibatinib', 'Abarelix', 'Sorafenib', 'Bosutinib', 'Axitinib', 'Ancestim', 'Alpelisib', 'Vemurafenib', 'Relugolix', 'Sunitinib', 'Selumetinib', 'Olaratumab', 'Histrelin', 'Imatinib', 'Ripretinib'}, 'G': {'Bosentan', 'Celecoxib', 'Danazol', 'Vemurafenib'}, 'J': {'Oseltamivir', 'Ampicillin'}, 'P': {'Atovaquone'}, 'A': {'Citric acid', 'Iron', 'Becaplermin'}, 'R': {'Doxapram'}, 'C': {'Heparin', 'Bosentan', 'Macitentan', 'Sitaxentan', 'Celecoxib', 'Pentaerythritol tetranitrate', 'Ambrisentan'}, 'H': {'Nafarelin', 'Linzagolix', 'Gonadorelin', 'Cetrorelix', 'Elagolix', 'Relugolix', 'Ganirelix'}}

In [3]:



Drugs with property L:

['Avapritinib', 'Regorafenib', 'Asciminib', 'Nilotinib', 'Sotorasib', 'Lenvatinib', 'Goserelin', 'Celecoxib', 'Triptorelin', 'Leuprolide', 'Degarelix', 'Pemigatinib', 'Glasdegib', 'Midostaurin', 'Cobimetinib', 'Encorafenib', 'Selpercatinib', 'Infigratinib', 'Leflunomide', 'Erdafitinib', 'Teriflunomide', 'Buserelin', 'Cisplatin', 'Trametinib', 'Vismodegib', 'Sonidegib', 'Tivozanib', 'Dabrafenib', 'Binimetinib', 'Pexidartinib', 'Futibatinib', 'Abarelix', 'Sorafenib', 'Bosutinib', 'Axitinib', 'Ancestim', 'Alpelisib', 'Vemurafenib', 'Relugolix', 'Sunitinib', 'Selumetinib', 'Olaratumab', 'Histrelin', 'Imatinib', 'Ripretinib']

In [5]:

Drugs not having property L:

['Darbepoetin alfa', 'Erythropoietin', 'Palifermin', 'Cetrorelix', 'Becaplermin', 'Oseltamivir', 'Ampicillin', 'Bosentan', 'Doxapram', 'Gonadorelin', 'Nafarelin', 'Hexachlorophene', 'Heparin', 'Atovaquone', 'Danazol', 'Iron', 'Citric acid', 'Roxadustat', 'Ingenol mebutate', 'Pentaerythritol tetranitrate', 'Lacosamide', 'Sitaxentan', 'Ambrisentan', 'Ganirelix', 'Halcinonide', 'Cholecystokinin', 'Peginesatide', 'Macitentan', 'Methoxy polyethylene glycol-epoetin beta', 'Nitrous acid', 'Aluminum chloride', 'Diboterminalfa', 'Elagolix', 'Ferrous gluconate', 'Ferrous succinate', 'Ferrous ascorbate', 'Ferrous fumarate', 'Ferrous glycine sulfate', 'Voxelator', 'Linzagolix', 'Darbepoetin alfa', 'Erythropoietin', 'Palifermin', 'Cetrorelix', 'Becaplermin', 'Oseltamivir', 'Ampicillin', 'Bosentan', 'Doxapram', 'Gonadorelin', 'Nafarelin', 'Hexachlorophene', 'Pentaerythritol tetranitrate', 'Heparin', 'Atovaquone', 'Danazol', 'Iron', 'Citric acid', 'Roxadustat', 'Ingenol mebutate', 'Lacosamide', 'Sitaxentan', 'Ambrisentan', 'Ganirelix', 'Halcinonide', 'Cholecystokinin', 'Peginesatide', 'Macitentan', 'Methoxy polyethylene glycol-epoetin beta', 'Nitrous acid', 'Aluminum chloride', 'Diboterminalfa', 'Elagolix', 'Ferrous gluconate', 'Ferrous succinate', 'Ferrous ascorbate', 'Ferrous fumarate', 'Ferrous glycine sulfate', 'Voxelator', 'Linzagolix']

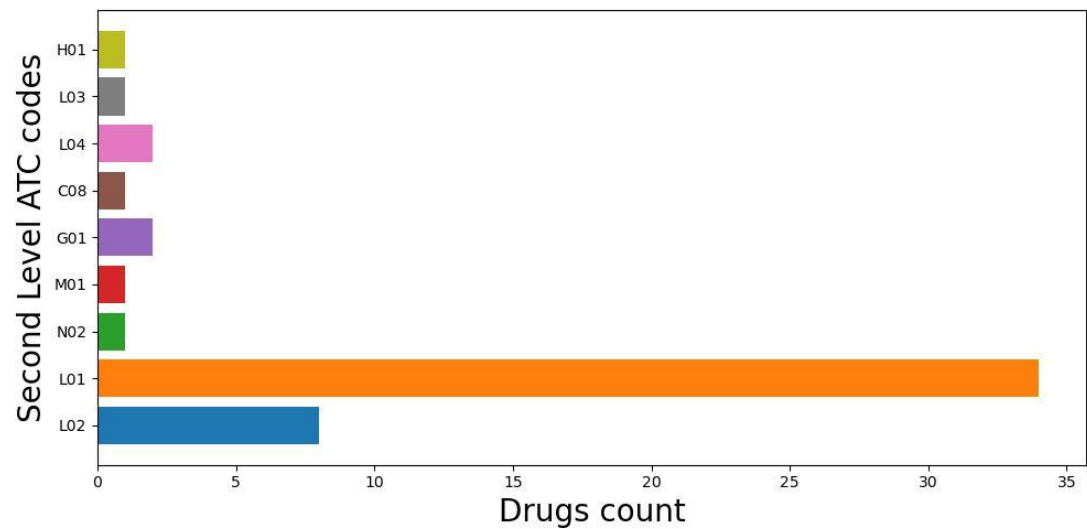
Accuracy: 52.94117647058823

Level 2 on L:

{'L02': {'Goserelin', 'Degarelix', 'Relugolix', 'Buserelin', 'Histrelin', 'Abarelix', 'Triptorelin', 'Leuprolide'}, 'L01': {'Avapritinib', 'Regorafenib', 'Asciminib', 'Nilotinib', 'Sotorasib', 'Lenvatinib', 'Celecoxib', 'Pemigatinib', 'Glasdegib', 'Midostaurin', 'Cobimetinib', 'Encorafenib', 'Selpercatinib', 'Infigratinib', 'Erdafitinib', 'Sonidegib', 'Cisplatin', 'Trametinib', 'Vismodegib', 'Tivozanib', 'Binimetinib', 'Dabrafenib', 'Pexidartinib', 'Futibatinib', 'Sorafenib', 'Bosutinib', 'Axitinib', 'Alpelisib', 'Vemurafenib', 'Sunitinib', 'Selumetinib', 'Olaratumab', 'Imatinib', 'Ripretinib'}, 'N02': {'Celecoxib'}, 'M01': {'Celecoxib'}, 'G01': {'Celecoxib', 'Vemurafenib'}, 'C08': {'Celecoxib'}, 'L04': {'Teriflunomide', 'Leflunomide'}, 'L03': {'Ancestim'}, 'H01': {'Relugolix'}}

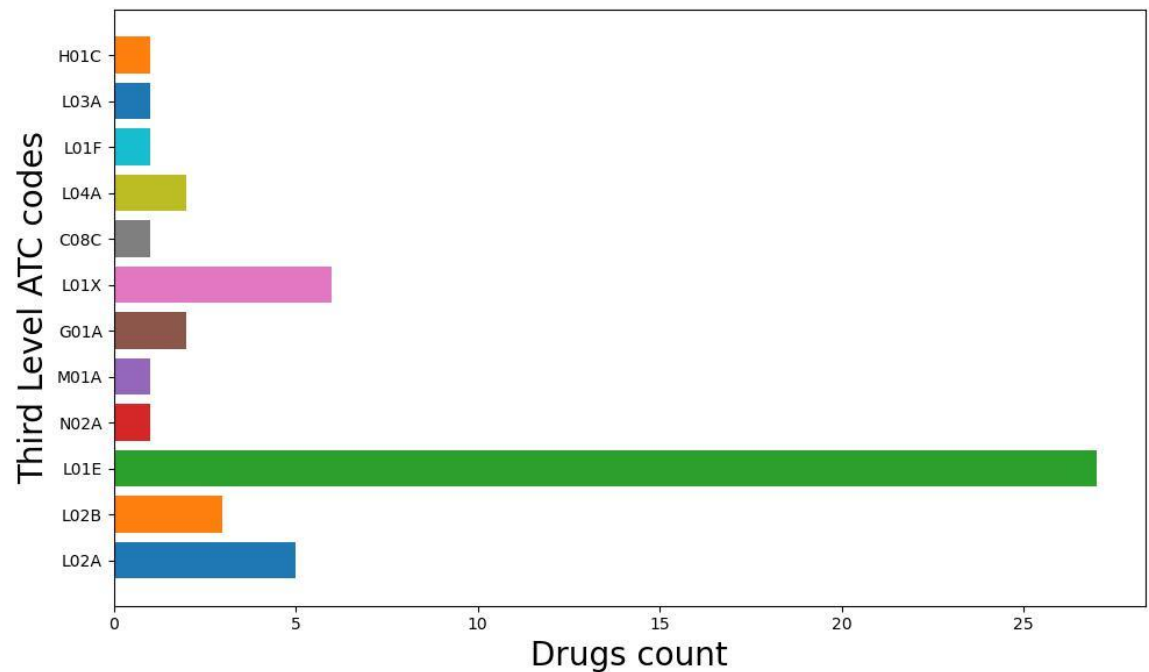
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Level 3 on L:

{'L02A': {'Buserelin', 'Histrelin', 'Goserelin', 'Triptorelin', 'Leuprolide'}, 'L02B': {'Relugolix', 'Abarelix', 'Degarelix'}, 'L01E': {'Avapritinib', 'Regorafenib', 'Asciminib', 'Nilotinib', 'Lenvatinib', 'Pemigatinib', 'Midostaurin', 'Cobimetinib', 'Encorafenib', 'Selpercatinib', 'Infigratinib', 'Erdafitinib', 'Pexidartinib', 'Trametinib', 'Tivozanib', 'Binimetinib', 'Dabrafenib', 'Futibatinib', 'Sorafenib', 'Bosutinib', 'Axitinib', 'Alpelisib', 'Vemurafenib', 'Sunitinib', 'Selumetinib', 'Imatinib', 'Ripretinib'}, 'N02A': {'Celecoxib'}, 'M01A': {'Celecoxib'}, 'G01A': {'Celecoxib', 'Vemurafenib'}, 'L01X': {'Sonidegib', 'Cisplatin', 'Sotorasib', 'Vismodegib', 'Glasdegib', 'Celecoxib'}, 'C08C': {'Celecoxib'}, 'L04A': {'Teriflunomide', 'Leflunomide'}, 'L01F': {'Olaratumab'}, 'L03A': {'Ancestim'}, 'H01C': {'Relugolix'}}

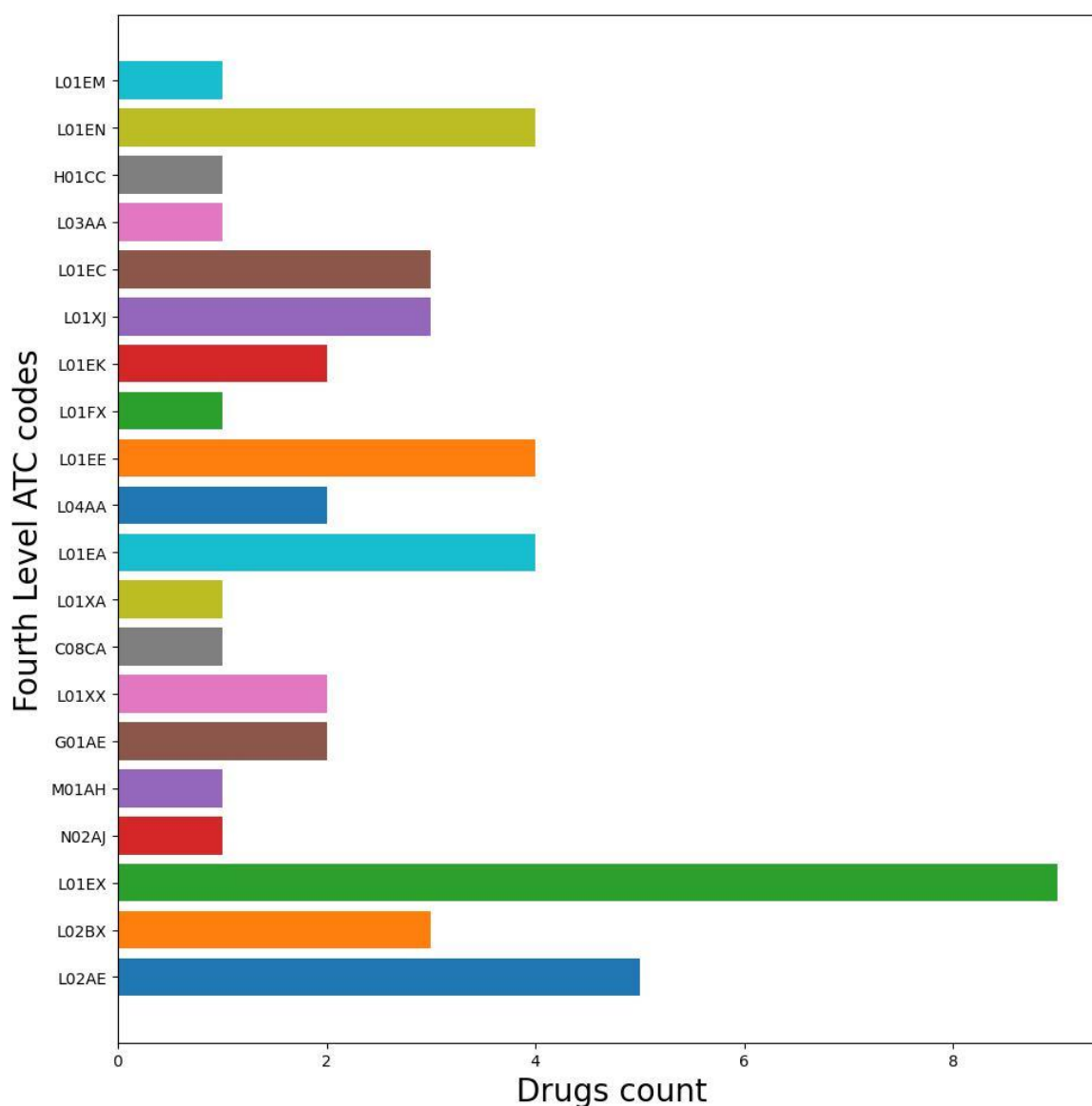


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Level 4 on L:

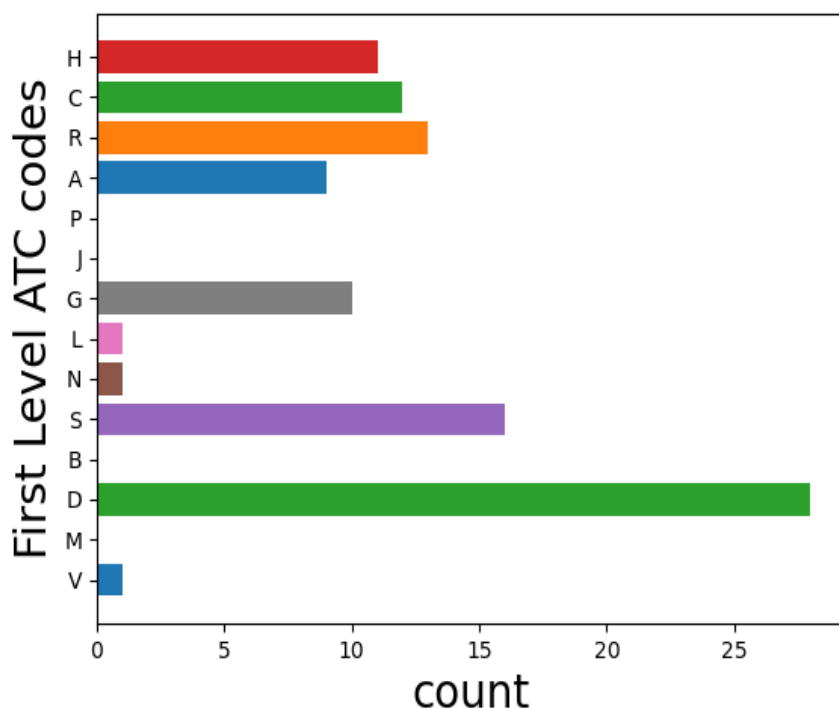
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Community/Cluster 5:

{'V': {'Prednisolone'}, 'M': set(), 'D': {'Amcinonide', 'Clocortolone', 'Prednisolone', 'Fluocinolone acetonide', 'Hydrocortisone probutate', 'Difluocortolone', 'Fluticasone', 'Betamethasone', 'Dexamethasone', 'Triamcinolone', 'Hydrocortisone butyrate', 'Prednicarbate', 'Flumethasone', 'Fluocinonide', 'Flurandrenolide', 'Desoximetasone', 'Diflorasone', 'Budesonide', 'Beclomethasone dipropionate', 'Mometasone furoate', 'Desonide', 'Fluorometholone', 'Hydrocortisone', 'Difluprednate', 'Clobetasone', 'Alclometasone', 'Ulobetasol', 'Methylprednisolone'}, 'B': set(), 'S': {'Flumethasone', 'Dexamethasone', 'Fludrocortisone', 'Hydrocortisone', 'Prednisolone', 'Rimexolone', 'Medrysone', 'Triamcinolone', 'Fluocinolone acetonide', 'Alclometasone', 'Clobetasone', 'Procaine', 'Desonide', 'Fluorometholone', 'Methylprednisolone', 'Betamethasone'}, 'N': {'Procaine'}, 'L': {'Megestrol acetate'}, 'G': {'Levonorgestrel', 'Drospirenone', 'Ulipristal', 'Progesterone', 'Megestrol acetate', 'Testosterone', 'Fluoxymesterone', 'Norethisterone', 'Mifepristone', 'Gestrinone'}, 'J': set(), 'P': set(), 'A': {'Tixocortol', 'Dexamethasone', 'Hydrocortisone', 'Betamethasone', 'Prednisolone', 'Triamcinolone', 'Budesonide', 'Beclomethasone dipropionate', 'Prednisone'}, 'R': {'Tixocortol', 'Dexamethasone', 'Hydrocortisone', 'Prednisolone', 'Fluticasone furoate', 'Triamcinolone', 'Budesonide', 'Beclomethasone dipropionate', 'Ciclesonide', 'Mometasone furoate', 'Fluticasone', 'Flunisolide', 'Betamethasone'}, 'C': {'Fluocinonide', 'Dexamethasone', 'Hydrocortisone', 'Finerenone', 'Prednisolone', 'Triamcinolone', 'Fluocinolone acetonide', 'Spironolactone', 'Procaine', 'Fluorometholone', 'Eplerenone', 'Betamethasone'}, 'H': {'Deflazacort', 'Dexamethasone', 'Fludrocortisone', 'Hydrocortisone', 'Betamethasone', 'Prednisolone', 'Rimexolone', 'Triamcinolone', 'Norethisterone', 'Methylprednisolone', 'Prednisone'}}

In [3]:



Drugs with property D:

['Amcinonide', 'Clocortolone', 'Prednisolone', 'Fluocinolone acetonide', 'Hydrocortisone probutate', 'Difluocortolone', 'Fluticasone', 'Betamethasone', 'Dexamethasone', 'Triamcinolone', 'Hydrocortisone butyrate', 'Prednicarbate', 'Flumethasone', 'Fluocinonide', 'Flurandrenolide', 'Desoximetasone', 'Diflorasone', 'Budesonide', 'Beclomethasone dipropionate', 'Mometasone furoate', 'Desonide', 'Fluorometholone', 'Hydrocortisone', 'Difluprednate', 'Clobetasone', 'Alclometasone', 'Ulobetasol', 'Methylprednisolone']

In [5]:

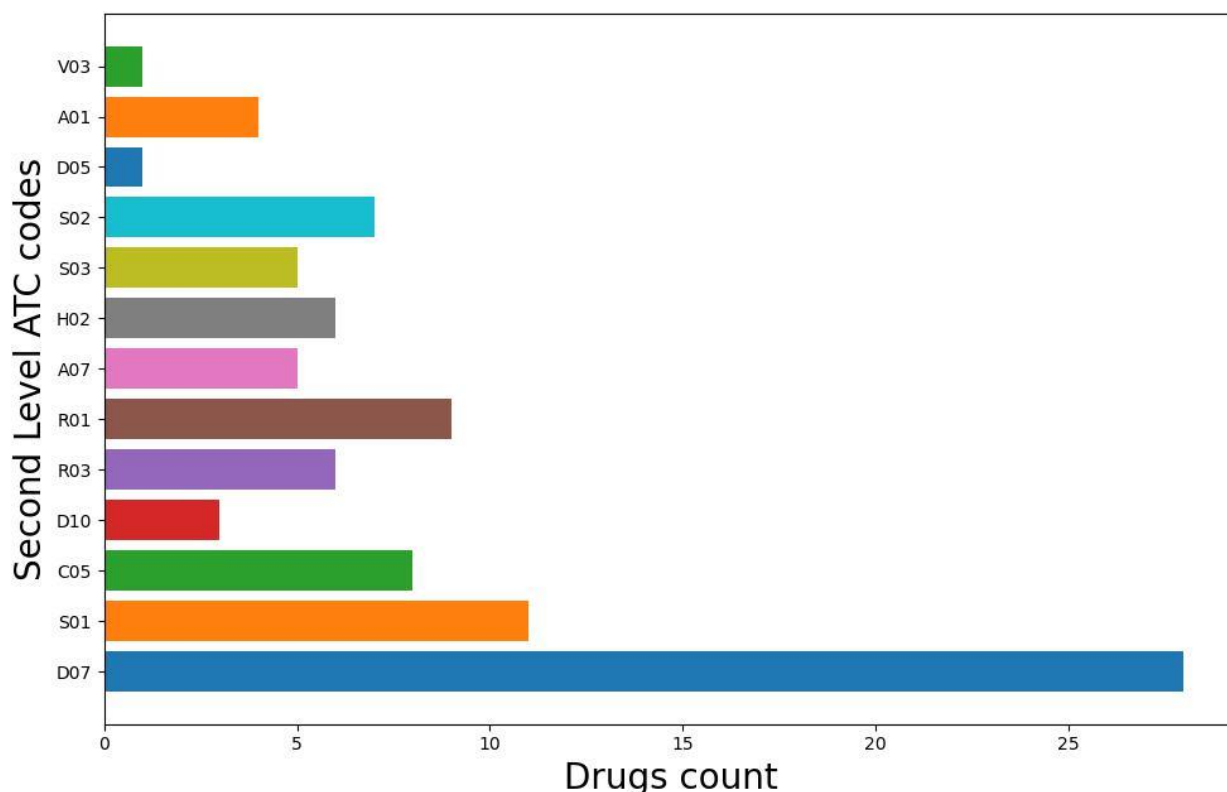
Drugs not having property D:

['Flunisolide', 'Medrysone', 'Megestrol acetate', 'Levonorgestrel', 'Progesterone', 'Spironolactone', 'Testosterone', 'Prednisone', 'Fludrocortisone', 'Eplerenone', 'Norethisterone', 'Procaine', 'Mifepristone', 'Rimexolone', 'Fluoxymesterone', 'Drospirenone', 'Ciclesonide', 'Ulipristal', 'Fluticasone furoate', 'Tixocortol', 'Gestrinone', 'Deflazacort', 'Finerenone', 'Flunisolide', 'Medrysone', 'Megestrol acetate', 'Levonorgestrel', 'Progesterone', 'Spironolactone', 'Testosterone', 'Prednisone', 'Fludrocortisone', 'Eplerenone', 'Norethisterone', 'Procaine', 'Mifepristone', 'Rimexolone', 'Fluoxymesterone', 'Drospirenone', 'Ciclesonide', 'Ulipristal', 'Fluticasone furoate', 'Tixocortol', 'Gestrinone', 'Deflazacort', 'Finerenone']

Accuracy: 54.90196078431372

Level 2 on D:

{'D07': {'Amcinonide', 'Clocortolone', 'Prednisolone', 'Fluocinolone acetonide', 'Hydrocortisone probutate', 'Difluocortolone', 'Fluticasone', 'Betamethasone', 'Dexamethasone', 'Triamcinolone', 'Hydrocortisone butyrate', 'Prednicarbate', 'Flumethasone', 'Fluocinonide', 'Flurandrenolide', 'Desoximetasone', 'Diflorasone', 'Budesonide', 'Beclomethasone dipropionate', 'Mometasone furoate', 'Desonide', 'Fluorometholone', 'Hydrocortisone', 'Difluprednate', 'Clobetasone', 'Alclometasone', 'Ulobetasol', 'Methylprednisolone'}, 'S01': {'Dexamethasone', 'Hydrocortisone', 'Prednisolone', 'Clobetasone', 'Triamcinolone', 'Fluocinolone acetonide', 'Alclometasone', 'Desonide', 'Fluorometholone', 'Methylprednisolone', 'Betamethasone'}, 'C05': {'Fluocinonide', 'Dexamethasone', 'Hydrocortisone', 'Prednisolone', 'Triamcinolone', 'Fluocinolone acetonide', 'Fluorometholone', 'Betamethasone'}, 'D10': {'Fluorometholone', 'Methylprednisolone', 'Dexamethasone'}, 'R03': {'Triamcinolone', 'Budesonide', 'Beclomethasone dipropionate', 'Mometasone furoate', 'Fluticasone', 'Betamethasone'}, 'R01': {'Dexamethasone', 'Hydrocortisone', 'Prednisolone', 'Triamcinolone', 'Budesonide', 'Beclomethasone dipropionate', 'Mometasone furoate', 'Fluticasone', 'Betamethasone'}, 'A07': {'Hydrocortisone', 'Prednisolone', 'Budesonide', 'Beclomethasone dipropionate', 'Betamethasone'}, 'H02': {'Dexamethasone', 'Hydrocortisone', 'Prednisolone', 'Triamcinolone', 'Methylprednisolone', 'Betamethasone'}, 'S03': {'Prednisolone', 'Dexamethasone', 'Hydrocortisone', 'Betamethasone'}, 'S02': {'Flumethasone', 'Dexamethasone', 'Hydrocortisone', 'Prednisolone', 'Triamcinolone', 'Fluocinolone acetonide', 'Betamethasone'}, 'D05': {'Ulobetasol'}, 'A01': {'Hydrocortisone', 'Prednisolone', 'Dexamethasone', 'Triamcinolone'}, 'V03': {'Prednisolone'}}



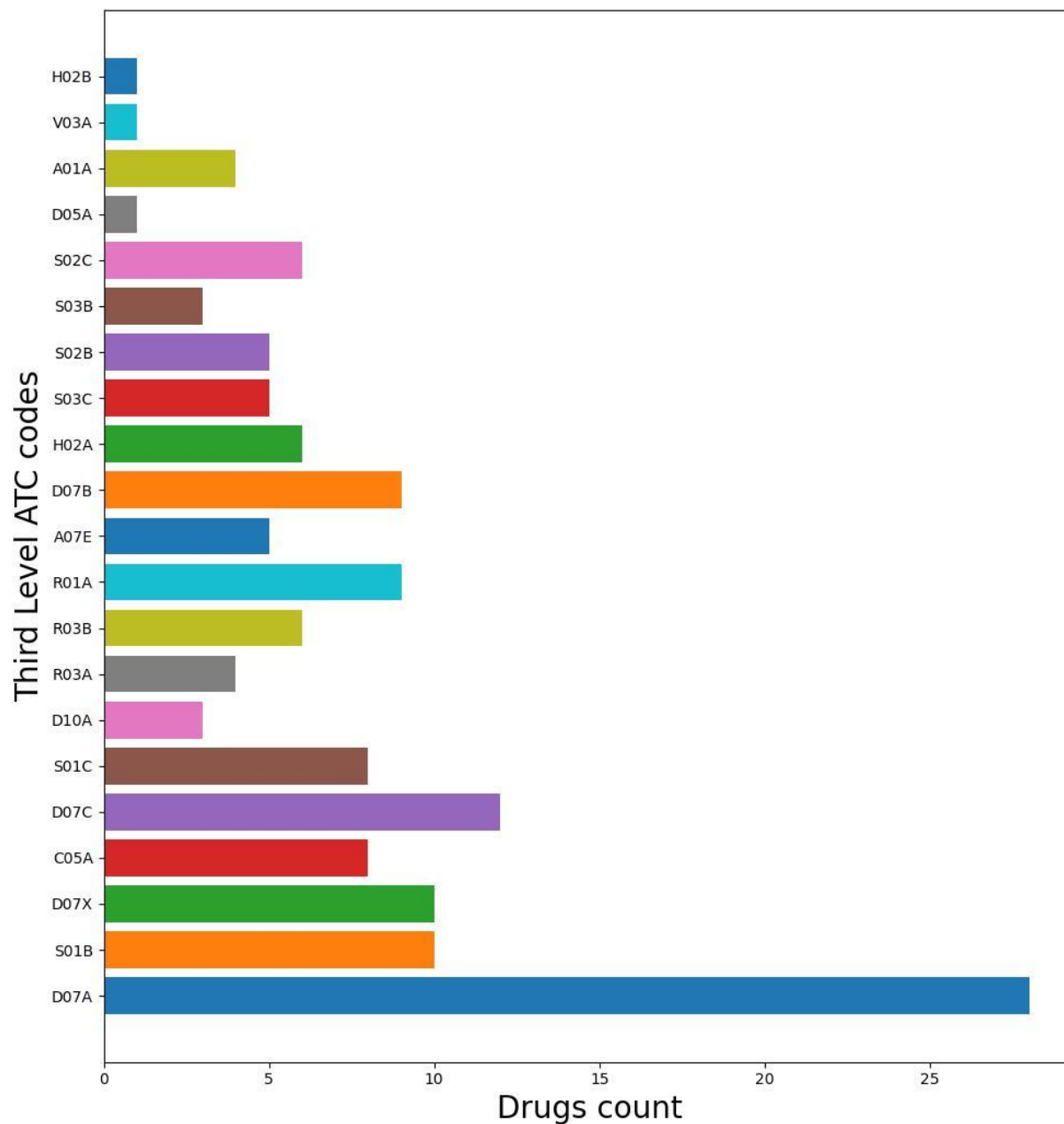
Level 3 on D:

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'Betamethasone'}, 'H02A': {'Dexamethasone', 'Hydrocortisone', 'Prednisolone', 'Triamcinolone', 'Methylprednisolone', 'Betamethasone'}, 'S03C': {'Prednisolone', 'Dexamethasone', 'Hydrocortisone', 'Betamethasone'}, 'S02B': {'Dexamethasone', 'Hydrocortisone', 'Prednisolone', 'Fluocinolone acetonide', 'Betamethasone'}, 'S03B': {'Prednisolone', 'Dexamethasone', 'Betamethasone'}, 'S02C': {'Flumethasone', 'Dexamethasone', 'Hydrocortisone', 'Prednisolone', 'Triamcinolone', 'Fluocinolone acetonide'}, 'D05A': {'Ulobetasol'}, 'A01A': {'Hydrocortisone', 'Prednisolone', 'Dexamethasone', 'Triamcinolone'}, 'V03A': {'Prednisolone'}, 'H02B': {'Methylprednisolone'}}

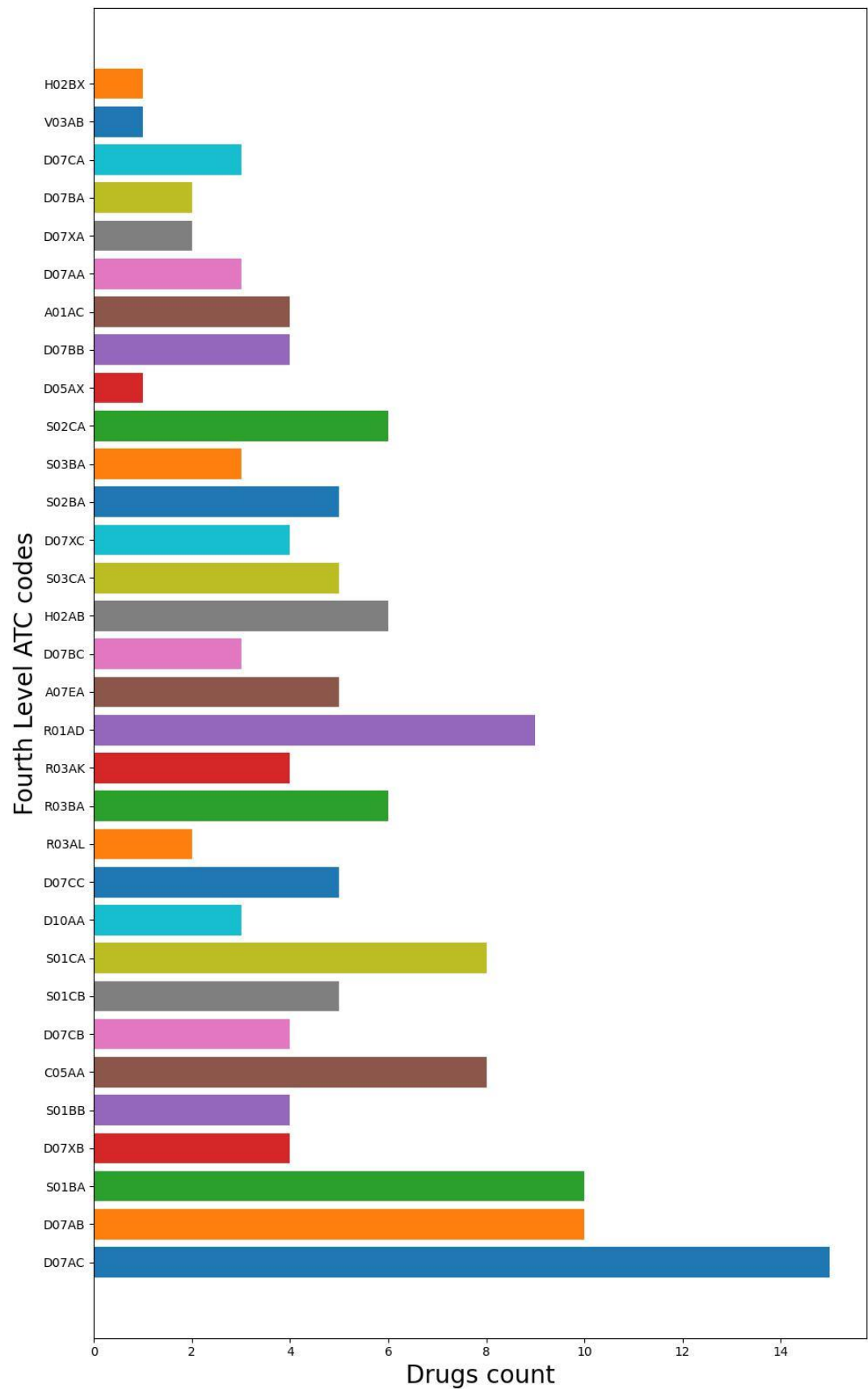


Level 4 on D:

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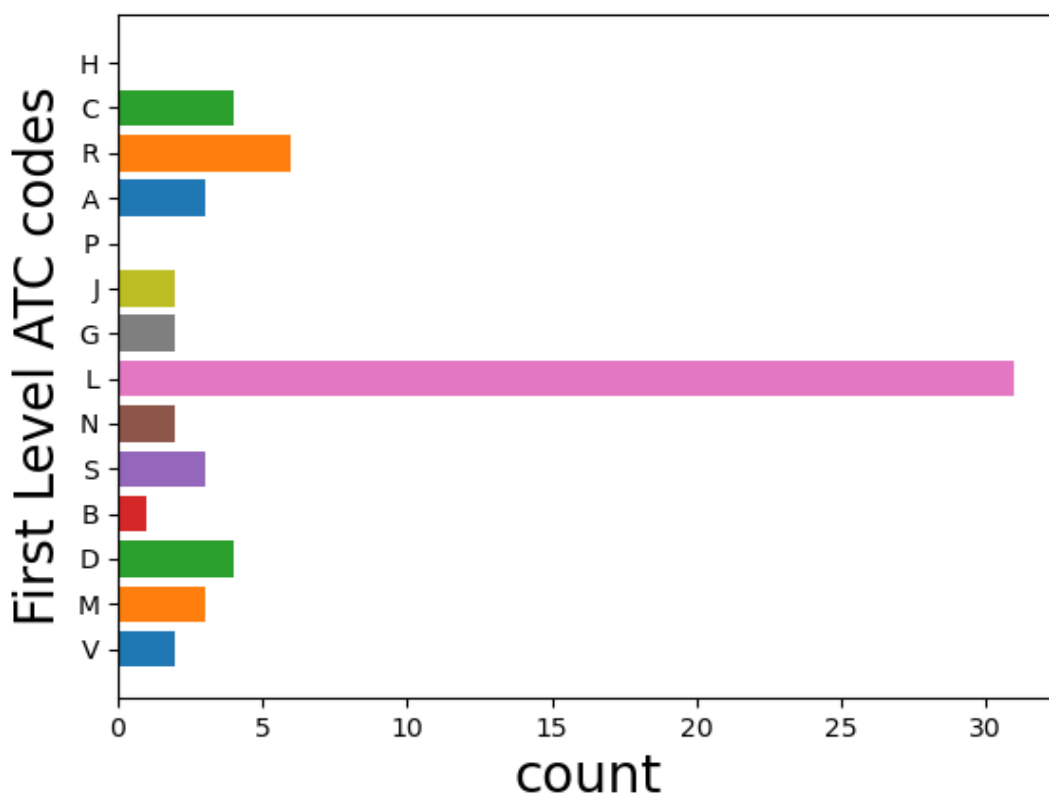
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Community/Cluster 6:

{'V': {'Acetylcysteine', 'Caffeine'}, 'M': {'Sulfinpyrazone', 'Auranofin', 'Dantrolene'}, 'D': {'Chloramphenicol', 'Abrocitinib', 'Caffeine', 'Tetracaine'}, 'B': {'Dipyridamole'}, 'S': {'Chloramphenicol', 'Acetylcysteine', 'Tetracaine'}, 'N': {'Caffeine', 'Tetracaine'}, 'L': {'Tofacitinib', 'Pentostatin', 'Acalabrutinib', 'Dasatinib', 'Streptozocin', 'Polatuzumab vedotin', 'Ipilimumab', 'Abatacept', 'Cladribine', 'Blinatumomab', 'Ibrutinib', 'Denileukin diftitox', 'Deucravacitinib', 'Tisagenlecleucel', 'Basiliximab', 'Catumaxomab', 'Baricitinib', 'Aldesleukin', 'Axicabtagene ciloleucel', 'Muromonab', 'Arsenic trioxide', 'Nintedanib', 'Ruxolitinib', 'Idelalisib', 'Pazopanib', 'Duvelisib', 'Zanubrutinib', 'Inebilizumab', 'Tafasitamab', 'Ponatinib', 'Copanlisib'}, 'G': {'Chloramphenicol', 'Bumetanide'}, 'J': {'Chloramphenicol', 'Didanosine'}, 'P': set(), 'A': {'Mesalazine', 'Crofelemer', 'Glyburide'}, 'R': {'Elexacaftor', 'Tezacaftor', 'Ivacaftor', 'Lumacaftor', 'Caffeine', 'Acetylcysteine'}, 'C': {'Triamterene', 'Amiloride', 'Bumetanide', 'Tetracaine'}, 'H': set()}

In [3]:



Drugs with property L:

['Tofacitinib', 'Pentostatin', 'Acalabrutinib', 'Dasatinib', 'Streptozocin', 'Polatuzumab vedotin', 'Ipilimumab', 'Abatacept', 'Cladribine', 'Blinatumomab', 'Ibrutinib', 'Denileukin diftitox', 'Deucravacitinib', 'Tisagenlecleucel', 'Basiliximab', 'Catumaxomab', 'Baricitinib', 'Aldesleukin', 'Axicabtagene ciloleucel', 'Muromonab', 'Arsenic trioxide', 'Nintedanib', 'Ruxolitinib', 'Idelalisib', 'Pazopanib', 'Duvelisib', 'Zanubrutinib', 'Inebilizumab', 'Tafasitamab', 'Ponatinib', 'Copanlisib']

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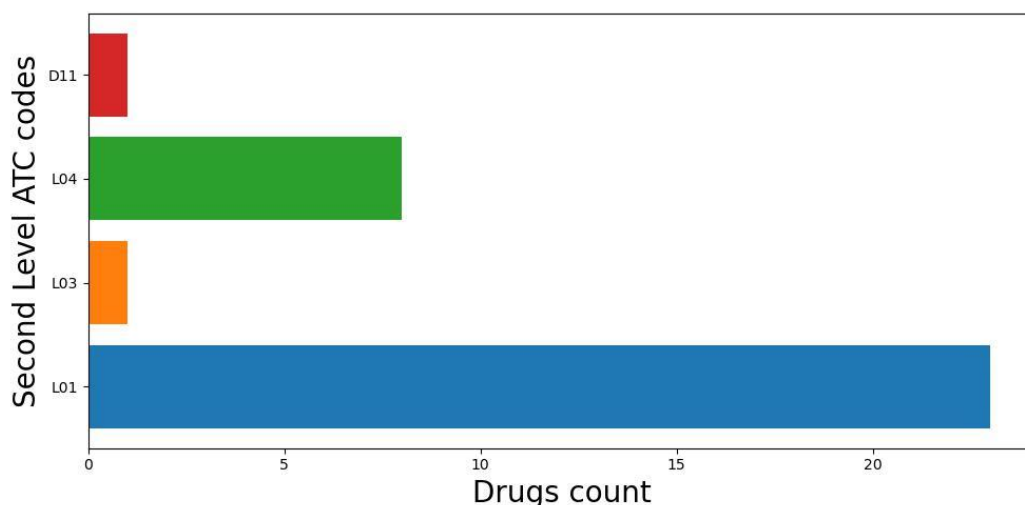
Drugs not having property L:

['Caffeine', 'Mesalazine', 'Triamterene', 'Chloramphenicol', 'Amiloride', 'Bumetanide', 'Didanosine', 'Dipyridamole', 'Auranofin', 'Glyburide', 'Sulfinpyrazone', 'Dantrolene', 'Crofelemer', 'Acetylcysteine', 'Ivacaftor', 'Tetracaine', 'Lumacaftor', 'Tezacaftor', 'Abrocitinib', 'Elexacaftor', 'Caffeine', 'Mesalazine', 'Triamterene', 'Chloramphenicol', 'Amiloride', 'Bumetanide', 'Didanosine', 'Dipyridamole', 'Auranofin', 'Glyburide', 'Sulfinpyrazone', 'Dantrolene', 'Crofelemer', 'Acetylcysteine', 'Ivacaftor', 'Tetracaine', 'Lumacaftor', 'Tezacaftor', 'Abrocitinib', 'Elexacaftor']

Accuracy: 60.7843137255

Level 2 on L:

{'L01': {'Pentostatin', 'Acalabrutinib', 'Dasatinib', 'Streptozocin', 'Polatuzumab vedotin', 'Ipilimumab', 'Cladribine', 'Blinatumomab', 'Ibrutinib', 'Denileukin diftitox', 'Tisagenlecleucel', 'Catumaxomab', 'Axicabtagene ciloleucel', 'Arsenic trioxide', 'Nintedanib', 'Ruxolitinib', 'Idelalisib', 'Pazopanib', 'Duvelisib', 'Zanubrutinib', 'Tafasitamab', 'Ponatinib', 'Copanlisib'}, 'L03': {'Aldesleukin'}, 'L04': {'Cladribine', 'Tofacitinib', 'Baricitinib', 'Muromonab', 'Deucravacitinib', 'Inebilizumab', 'Basiliximab', 'Abatacept'}}

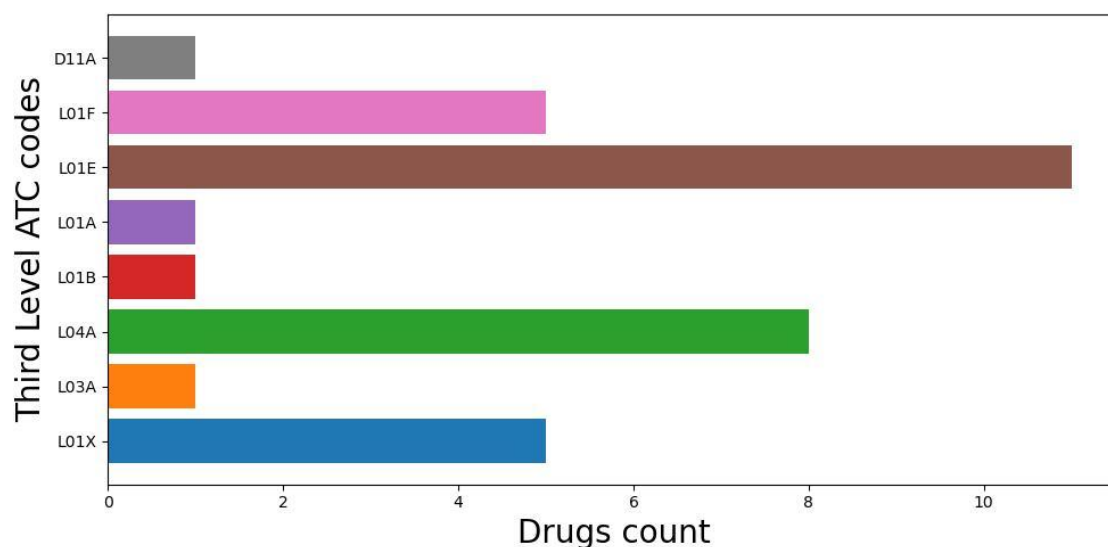


Level 3 on L:

{'L01X': {'Pentostatin', 'Axicabtagene ciloleucel', 'Denileukin diftitox', 'Tisagenlecleucel', 'Arsenic trioxide'}, 'L03A': {'Aldesleukin'}, 'L04A': {'Cladribine', 'Tofacitinib', 'Baricitinib', 'Muromonab', 'Deucravacitinib', 'Inebilizumab', 'Basiliximab', 'Abatacept'}, 'L01B': {'Cladribine'}, 'L01A': {'Streptozocin'}, 'L01E': {'Pazopanib', 'Duvelisib', 'Zanubrutinib', 'Ibrutinib', 'Acalabrutinib', 'Dasatinib', 'Nintedanib', 'Ponatinib', 'Copanlisib', 'Idelalisib', 'Ruxolitinib'}, 'L01F': {'Catumaxomab', 'Blinatumomab', 'Tafasitamab', 'Polatuzumab vedotin', 'Ipilimumab'}}

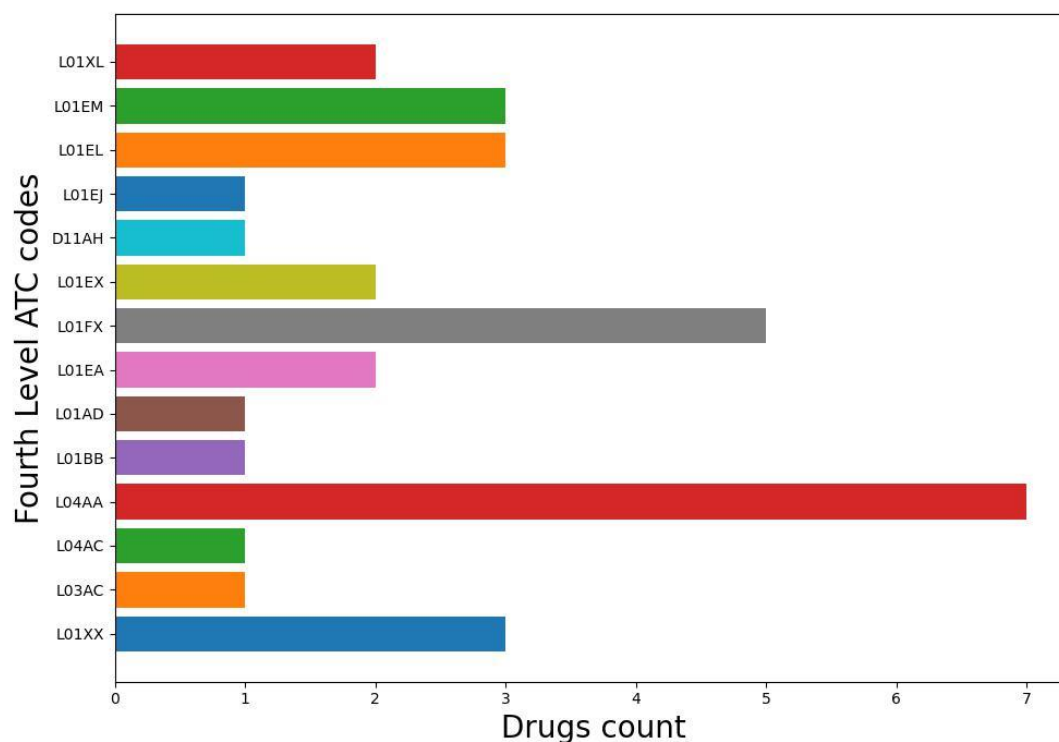
Drug Repositioning Pipeline Integrating Community Analysis in Drug-Drug Similarity Networks and Automated ATC Community Labeling to Enhance Molecular Docking Validation

D. Colibășanu, V. Groza, M.A. Occhiuzzi, F. Grande, M. Udrescu, L. Udrescu



Level 4 on L:

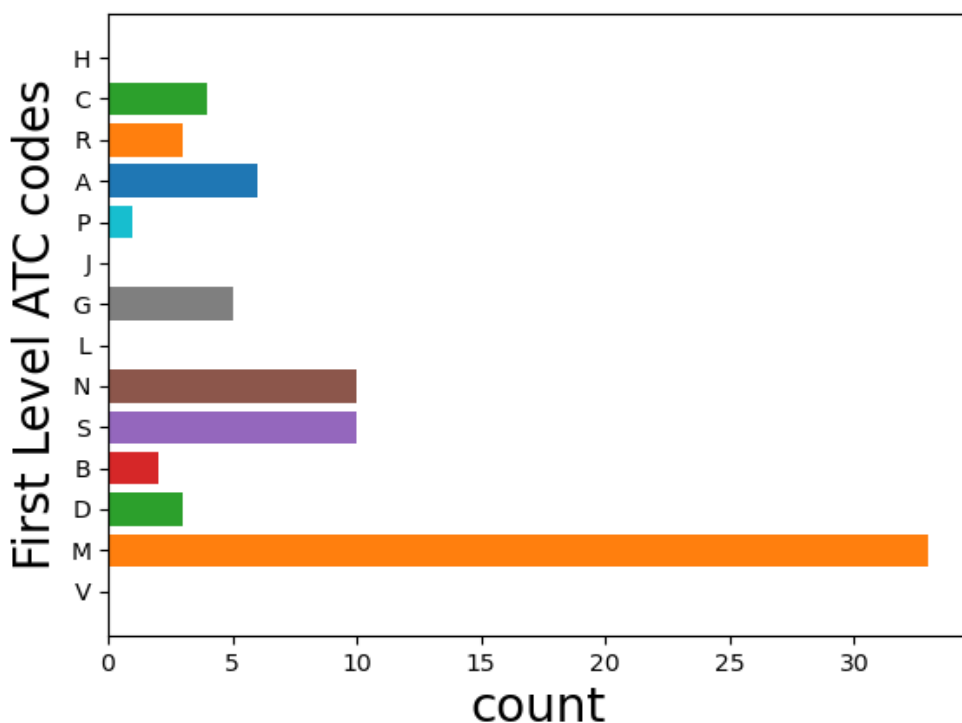
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Community/Cluster 7:

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{'V': set(), 'M': {'Aceclofenac', 'Dexketoprofen', 'Sulindac', 'Lumiracoxib', 'Piroxicam', 'Naproxen', 'Bendazac', 'Acemetacin', 'Tenoxicam', 'Flurbiprofen', 'Ibuprofen', 'Etodolac', 'Phenylbutazone', 'Nabumetone', 'Oxaprozin', 'Tiaprofenic acid', 'Lornoxicam', 'Loxoprofen', 'Tolmetin', 'Ketorolac', 'Tolfenamic acid', 'Meloxicam', 'Bufexamac', 'Ketoprofen', 'Dexibuprofen', 'Suprofen', 'Acetylsalicylic acid', 'Indomethacin', 'Fenoprofen', 'Flufenamic acid', 'Diclofenac', 'Meclofenamic acid', 'Mefenamic acid'}, 'D': {'Salicylic acid', 'Minoxidil', 'Diclofenac'}, 'B': {'Acetylsalicylic acid', 'Triflusal'}, 'S': {'Flurbiprofen', 'Salicylic acid', 'Antipyrine', 'Nepafenac', 'Bromfenac', 'Ketorolac', 'Indomethacin', 'Piroxicam', 'Bendazac', 'Diclofenac'}, 'N': {'Meloxicam', 'Salicylic acid', 'Ibuprofen', 'Salsalate', 'Antipyrine', 'Acetaminophen', 'Metamizole', 'Dexketoprofen', 'Diflunisal', 'Acetylsalicylic acid'}, 'L': set(), 'G': {'Sulfasalazine', 'Phenyl salicylate', 'Ibuprofen', 'Papaverine', 'Naproxen'}, 'J': set(), 'P': {'Diethylcarbamazine'}, 'A': {'Sulfasalazine', 'Troglitazone', 'Rosiglitazone', 'Papaverine', 'Acetylsalicylic acid', 'Balsalazide'}, 'R': {'Flurbiprofen', 'Ibuprofen', 'Pseudoephedrine'}, 'C': {'Acetylsalicylic acid', 'Ibuprofen', 'Minoxidil', 'Indomethacin'}, 'H': set()}
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In [3]:



Drugs with property M:

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['Aceclofenac', 'Dexketoprofen', 'Sulindac', 'Lumiracoxib', 'Piroxicam', 'Naproxen', 'Bendazac', 'Acemetacin', 'Tenoxicam', 'Flurbiprofen', 'Ibuprofen', 'Etodolac', 'Phenylbutazone', 'Nabumetone', 'Oxaprozin', 'Tiaprofenic acid', 'Lornoxicam', 'Loxoprofen', 'Tolmetin', 'Ketorolac', 'Tolfenamic acid', 'Meloxicam', 'Bufexamac', 'Ketoprofen', 'Dexibuprofen', 'Suprofen', 'Acetylsalicylic acid', 'Indomethacin', 'Fenoprofen', 'Flufenamic acid', 'Diclofenac', 'Meclofenamic acid', 'Mefenamic acid']
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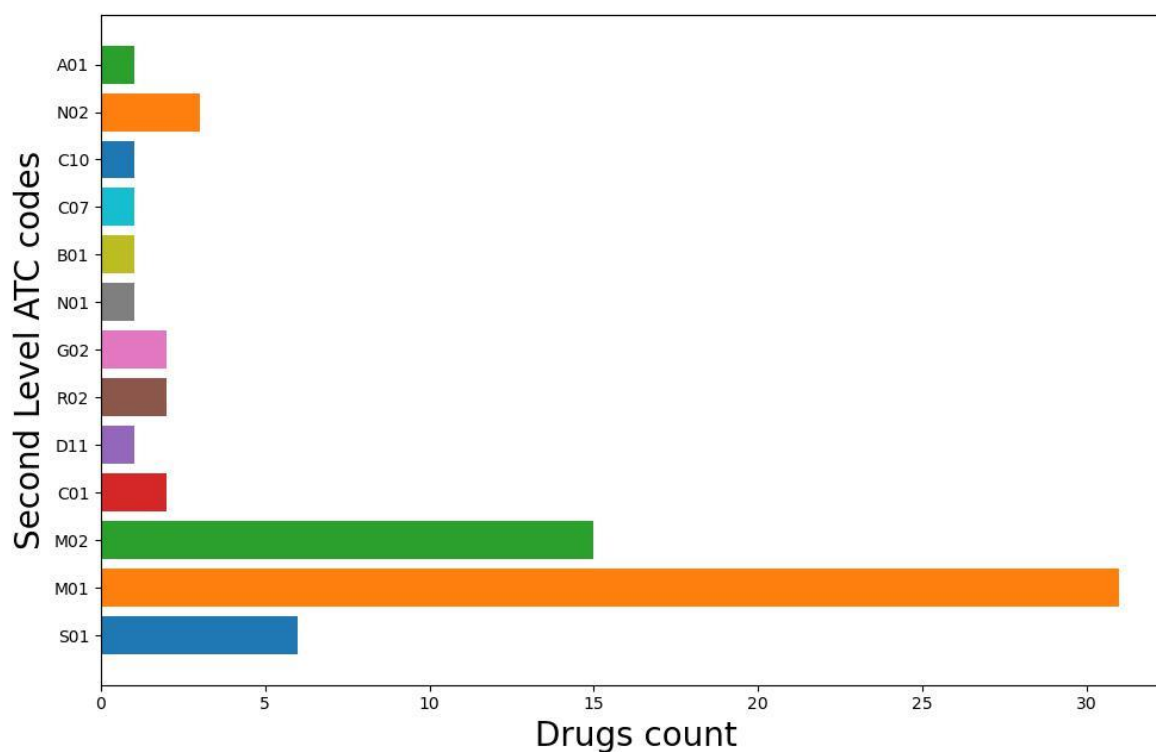
Drugs not having property M:

['Troglitazone', 'Acetaminophen', 'Minoxidil', 'Rosiglitazone', 'Diethylcarbamazine', 'Sulfasalazine', 'Pseudoephedrine', 'Diflunisal', 'Salicylic acid', 'Bromfenac', 'Balsalazide', 'Papaverine', 'Salsalate', 'Antipyrine', 'Metamizole', 'Nepafenac', 'Triflusal', 'Phenyl salicylate', 'Troglitazone', 'Acetaminophen', 'Minoxidil', 'Rosiglitazone', 'Diethylcarbamazine', 'Sulfasalazine', 'Pseudoephedrine', 'Diflunisal', 'Salicylic acid', 'Bromfenac', 'Balsalazide', 'Papaverine', 'Salsalate', 'Antipyrine', 'Metamizole', 'Nepafenac', 'Triflusal', 'Phenyl salicylate']

Accuracy: 64.70588235294117

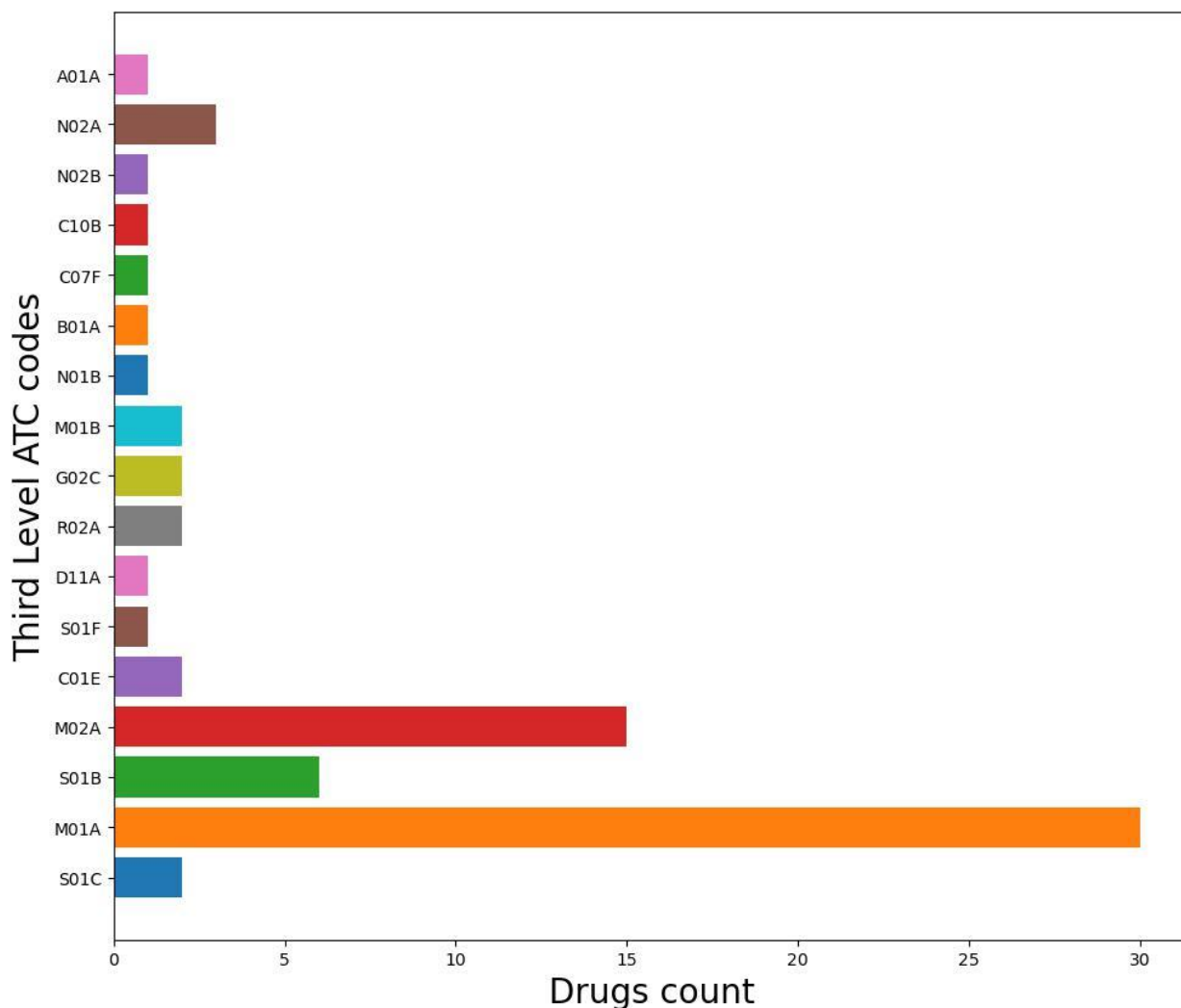
Level 2 on M:

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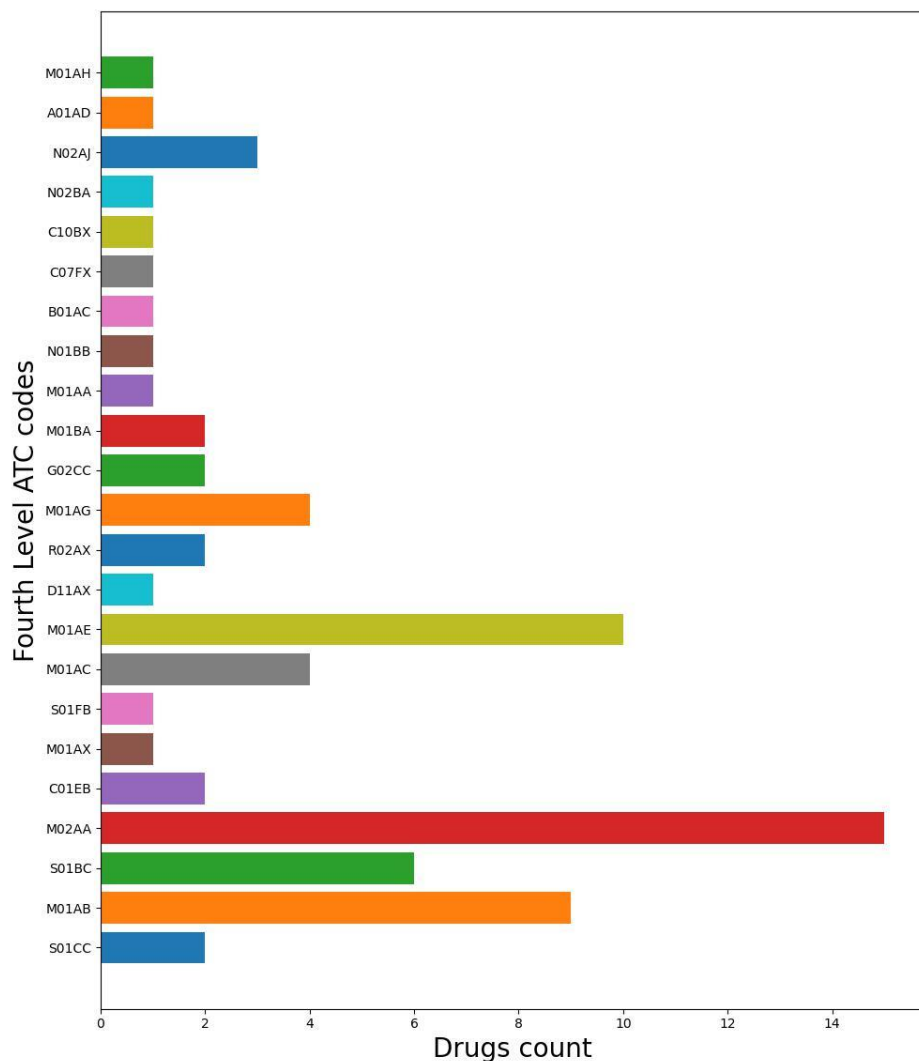
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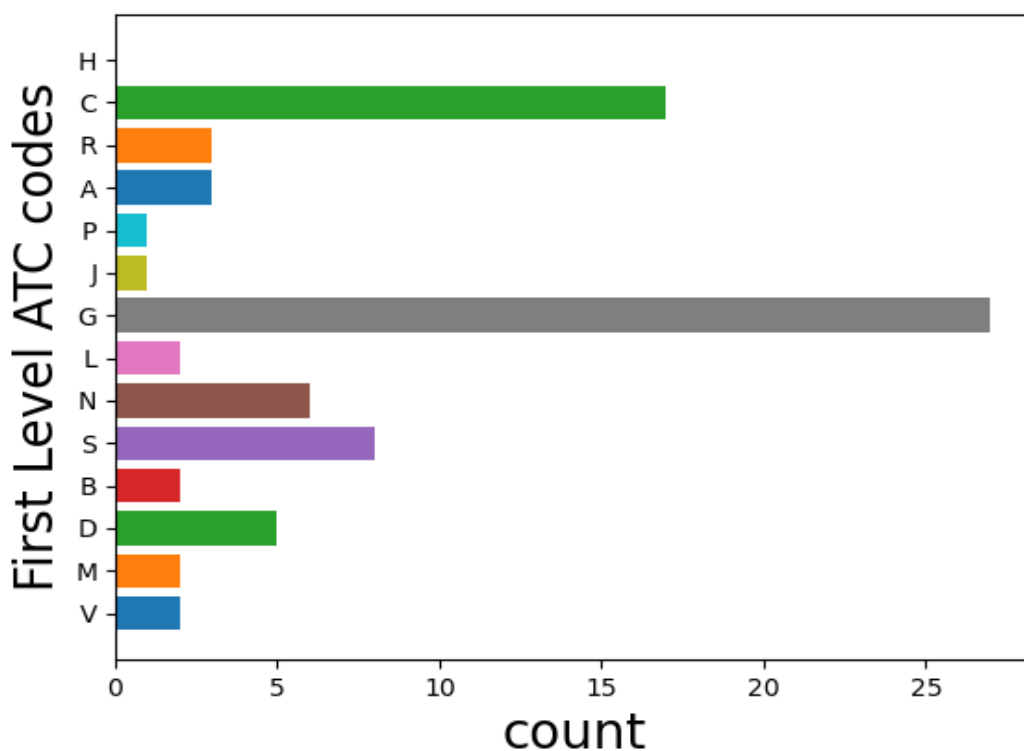
Level 4 on M:

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Community/Cluster 8:

{'V': {'Vitamin A', 'Diazoxide'}, 'M': {'Hyaluronic acid', 'Valdecocib'}, 'D': {'Vitamin A', 'Hyaluronic acid', 'Tretinoin', 'Urea', 'Dequalinium'}, 'B': {'Urea', 'Potassium chloride'}, 'S': {'Methazolamide', 'Vitamin A', 'Diclofenamide', 'Hyaluronic acid', 'Brinzolamide', 'Dorzolamide', 'Acetazolamide', 'Voretigene neparvovec'}, 'N': {'Zonisamide', 'Sulpiride', 'Nitrazepam', 'Sulthiame', 'Tramadol', 'Phenacemide'}, 'L': {'Mycophenolic acid', 'Tretinoin'}, 'G': {'Diclofenamide', 'Zonisamide', 'Metolazone', 'Hydroflumethiazide', 'Polythiazide', 'Metergoline', 'Dorzolamide', 'Trichlormethiazide', 'Quinethazone', 'Hydrochlorothiazide', 'Diazoxide', 'Acetazolamide', 'Piretanide', 'Chlorthalidone', 'Indapamide', 'Vardenafil', 'Cyclothiazide', 'Torasemide', 'Tadalafil', 'Phenazopyridine', 'Bendroflumethiazide', 'Methazolamide', 'Brinzolamide', 'Chlorothiazide', 'Furosemide', 'Dequalinium', 'Benzthiazide'}, 'J': {'Ribavirin'}, 'P': {'Permethrin'}, 'A': {'Vitamin A', 'Sodium sulfate', 'Potassium chloride'}, 'R': {'Dequalinium', 'Vitamin A', 'Hyaluronic acid'}, 'C': {'Indapamide', 'Hydrochlorothiazide', 'Xanthinol', 'Metolazone', 'Cyclothiazide', 'Torasemide', 'Hydroflumethiazide', 'Polythiazide', 'Diazoxide', 'Tadalafil', 'Trichlormethiazide', 'Chlorothiazide', 'Furosemide', 'Quinethazone', 'Bendroflumethiazide', 'Piretanide', 'Chlorthalidone'}, 'H': set() }



Drugs with property G:

['Diclofenamide', 'Zonisamide', 'Metolazone', 'Hydroflumethiazide', 'Polythiazide', 'Metergoline', 'Dorzolamide', 'Trichlormethiazide', 'Quinethazone', 'Hydrochlorothiazide', 'Diazoxide', 'Acetazolamide', 'Piretanide', 'Chlorthalidone', 'Indapamide', 'Vardenafil', 'Cyclothiazide', 'Torasemide', 'Tadalafil', 'Phenazopyridine', 'Bendroflumethiazide', 'Methazolamide', 'Brinzolamide', 'Chlorothiazide', 'Furosemide', 'Dequalinium', 'Benzthiazide']

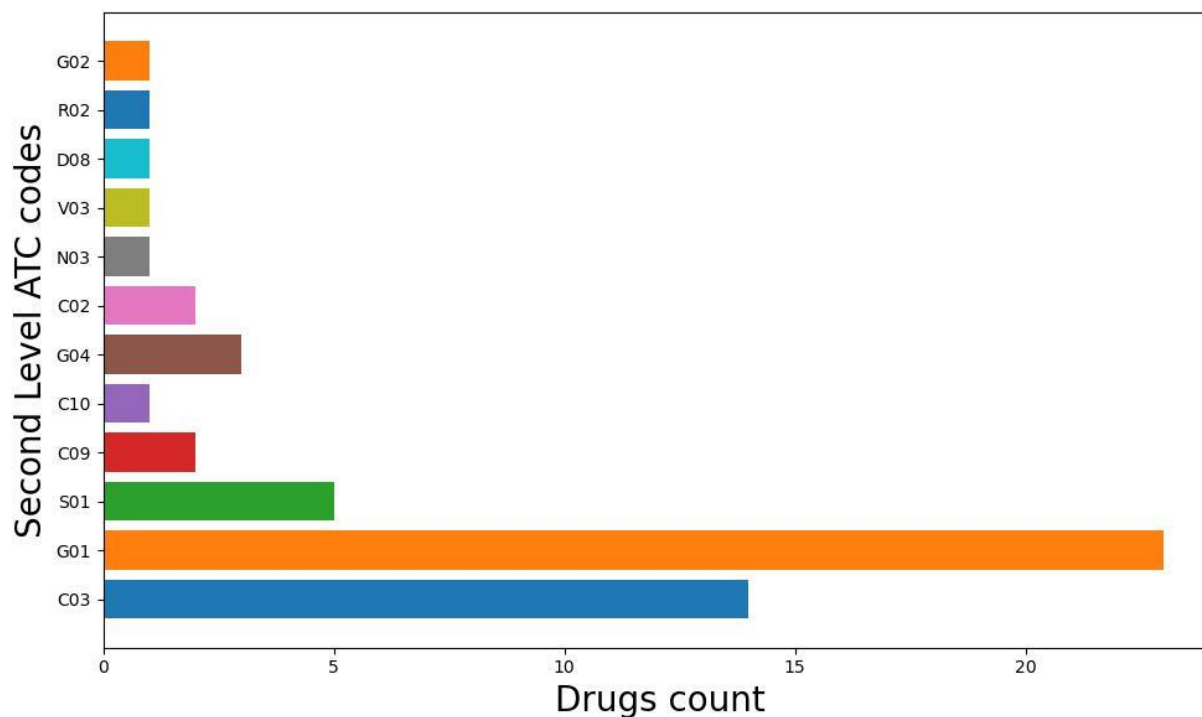
Drugs not having property G:

['Vitamin A', 'Tramadol', 'Sulpiride', 'Valdecoxib', 'Tretinoin', 'Potassium chloride', 'Ribavirin', 'Mycophenolic acid', 'Phenacemide', 'Nitrazepam', 'Urea', 'Permethrin', 'Sulthiame', 'Hyaluronic acid', 'Xanthinol', 'Sodium sulfate', 'Voretigene neparovec', 'Vitamin A', 'Tramadol', 'Sulpiride', 'Valdecoxib', 'Tretinoin', 'Potassium chloride', 'Ribavirin', 'Mycophenolic acid', 'Phenacemide', 'Nitrazepam', 'Urea', 'Permethrin', 'Sulthiame', 'Hyaluronic acid', 'Xanthinol', 'Sodium sulfate', 'Voretigene neparovec']

Accuracy: 61.36363636363637

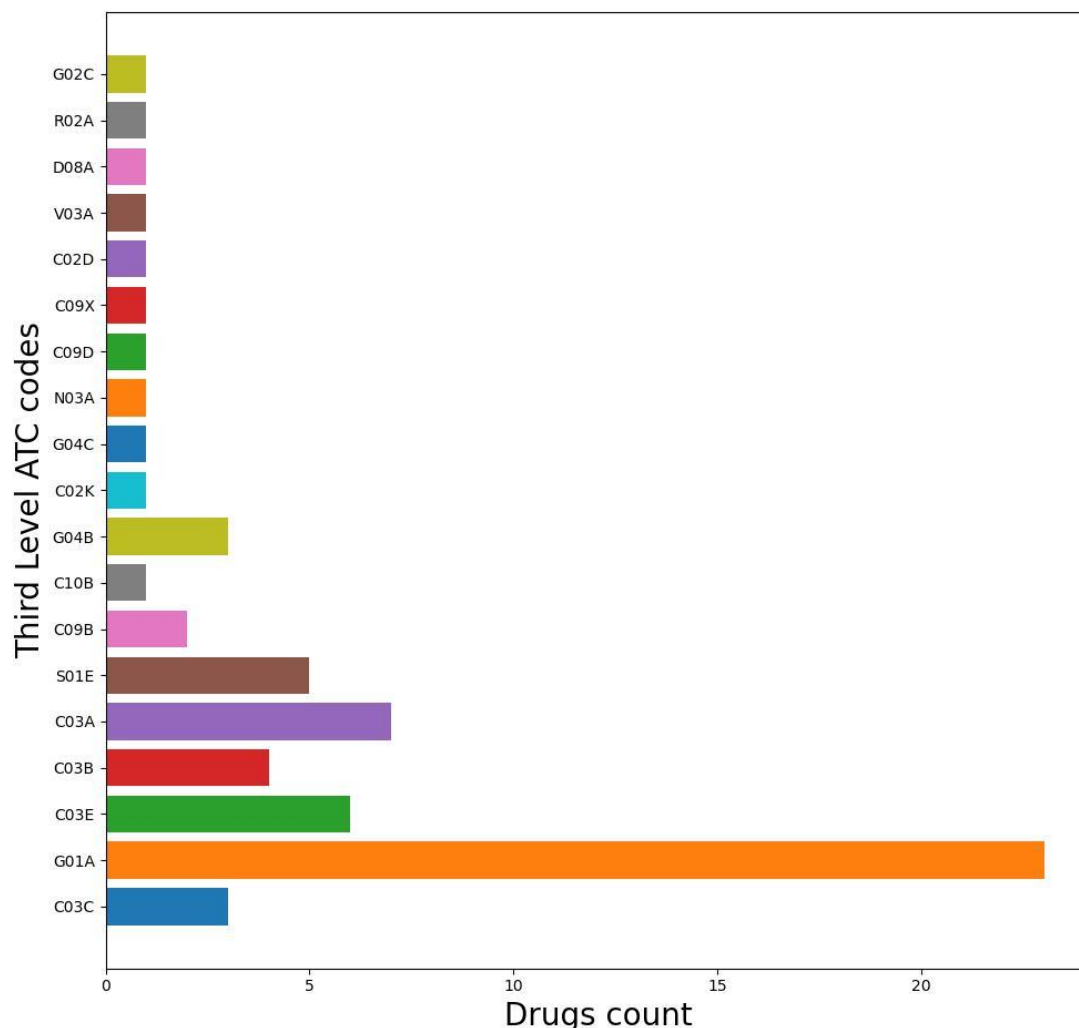
Level 2 on G:

{'C03': {'Indapamide', 'Hydrochlorothiazide', 'Metolazone', 'Cyclothiazide', 'Torasemide', 'Hydroflumethiazide', 'Polythiazide', 'Trichlormethiazide', 'Chlorothiazide', 'Furosemide', 'Quinethazone', 'Bendroflumethiazide', 'Piretanide', 'Chlorthalidone'}, 'G01': {'Diclofenamide', 'Zonisamide', 'Metolazone', 'Hydroflumethiazide', 'Polythiazide', 'Dorzolamide', 'Trichlormethiazide', 'Quinethazone', 'Hydrochlorothiazide', 'Diazoxide', 'Acetazolamide', 'Piretanide', 'Chlorthalidone', 'Indapamide', 'Cyclothiazide', 'Torasemide', 'Bendroflumethiazide', 'Methazolamide', 'Brinzolamide', 'Chlorothiazide', 'Furosemide', 'Dequalinium', 'Benzthiazide'}, 'S01': {'Methazolamide', 'Diclofenamide', 'Brinzolamide', 'Dorzolamide', 'Acetazolamide'}, 'C09': {'Indapamide', 'Hydrochlorothiazide'}, 'C10': {'Indapamide'}, 'G04': {'Tadalafil', 'Phenazopyridine', 'Vardenafil'}, 'C02': {'Tadalafil', 'Diazoxide'}, 'N03': {'Zonisamide'}, 'V03': {'Diazoxide'}, 'D08': {'Dequalinium'}, 'R02': {'Dequalinium'}, 'G02': {'Metergoline'}}



Level 3 on G:

{'C03C': {'Torasemide', 'Piretanide', 'Furosemide'}, 'G01A': {'Diclofenamide', 'Zonisamide', 'Metolazone', 'Hydroflumethiazide', 'Polythiazide', 'Dorzolamide', 'Trichlormethiazide', 'Quinethazone', 'Hydrochlorothiazide', 'Diazoxide', 'Acetazolamide', 'Piretanide', 'Chlorthalidone', 'Indapamide', 'Cyclothiazide', 'Torasemide', 'Bendroflumethiazide', 'Methazolamide', 'Brinzolamide', 'Chlorothiazide', 'Furosemide', 'Dequalinium', 'Benzthiazide'}, 'C03E': {'Hydrochlorothiazide', 'Metolazone', 'Trichlormethiazide', 'Furosemide', 'Bendroflumethiazide', 'Chlorthalidone'}, 'C03B': {'Indapamide', 'Quinethazone', 'Chlorthalidone', 'Metolazone'}, 'C03A': {'Hydrochlorothiazide', 'Cyclothiazide', 'Hydroflumethiazide', 'Polythiazide', 'Trichlormethiazide', 'Chlorothiazide', 'Bendroflumethiazide'}, 'S01E': {'Methazolamide', 'Diclofenamide', 'Brinzolamide', 'Dorzolamide', 'Acetazolamide'}, 'C09B': {'Indapamide', 'Hydrochlorothiazide'}, 'C10B': {'Indapamide'}, 'G04B': {'Tadalafil', 'Phenazopyridine', 'Vardenafil'}, 'C02K': {'Tadalafil'}, 'G04C': {'Tadalafil'}, 'N03A': {'Zonisamide'}, 'C09D': {'Hydrochlorothiazide'}, 'C09X': {'Hydrochlorothiazide'}, 'C02D': {'Diazoxide'}, 'V03A': {'Diazoxide'}, 'D08A': {'Dequalinium'}, 'R02A': {'Dequalinium'}, 'G02C': {'Metergoline'}}

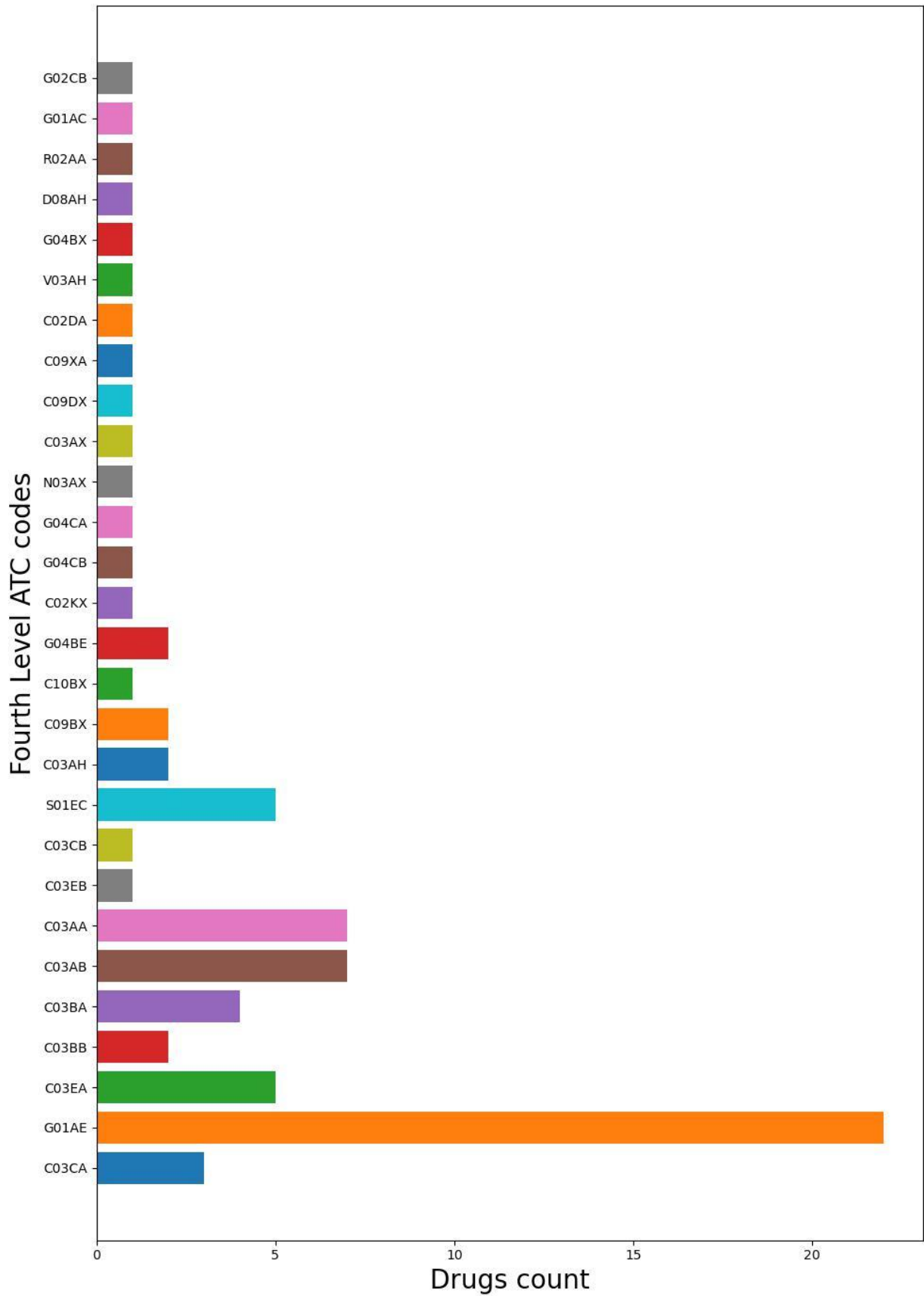


Level 4 on G:

{'C03CA': {'Torasemide', 'Piretanide', 'Furosemide'}, 'G01AE': {'Diclofenamide', 'Zonisamide', 'Metolazone', 'Hydroflumethiazide', 'Polythiazide', 'Dorzolamide', 'Trichlormethiazide', 'Quinethazone', 'Hydrochlorothiazide', 'Diazoxide', 'Acetazolamide', 'Piretanide', 'Chlorthalidone', 'Indapamide', 'Cyclothiazide', 'Torasemide', 'Bendroflumethiazide', 'Methazolamide', 'Brinzolamide', 'Chlorothiazide', 'Furosemide', 'Benzthiazide'}, 'C03EA': {'Hydrochlorothiazide', 'Metolazone', 'Trichlormethiazide', 'Bendroflumethiazide', 'Chlorthalidone'}, 'C03BB': {'Quinethazone', 'Chlorthalidone'}, 'C03BA': {'Indapamide', 'Quinethazone', 'Chlorthalidone', 'Metolazone'}, 'C03AB': {'Hydrochlorothiazide', 'Cyclothiazide', 'Hydroflumethiazide', 'Polythiazide', 'Trichlormethiazide', 'Chlorothiazide', 'Bendroflumethiazide'}, 'C03AA': {'Hydrochlorothiazide', 'Cyclothiazide', 'Hydroflumethiazide', 'Polythiazide', 'Trichlormethiazide', 'Chlorothiazide', 'Bendroflumethiazide'}, 'C03EB': {'Furosemide'}, 'C03CB': {'Furosemide'}, 'S01EC': {'Methazolamide', 'Diclofenamide', 'Brinzolamide', 'Dorzolamide', 'Acetazolamide'}, 'C03AH': {'Hydroflumethiazide', 'Chlorothiazide'}, 'C09BX': {'Indapamide', 'Hydrochlorothiazide'}, 'C10BX': {'Indapamide'}, 'G04BE': {'Tadalafil', 'Vardenafil'}, 'C02KX': {'Tadalafil'}, 'G04CA': {'Tadalafil'}, 'N03AX': {'Zonisamide'}, 'C03AX': {'Hydrochlorothiazide'}, 'C09DX': {'Hydrochlorothiazide'}, 'C09XA': {'Hydrochlorothiazide'}, 'C02DA': {'Diazoxide'}, 'V03AH': {'Diazoxide'}, 'G04BX': {'Phenazopyridine'}, 'D08AH': {'Dequalinium'}, 'R02AA': {'Dequalinium'}, 'G01AC': {'Dequalinium'}, 'G02CB': {'Metergoline'}}

Drug Repositioning Pipeline Integrating Community Analysis in Drug-Drug Similarity Networks and Automated ATC Community Labeling to Enhance Molecular Docking Validation

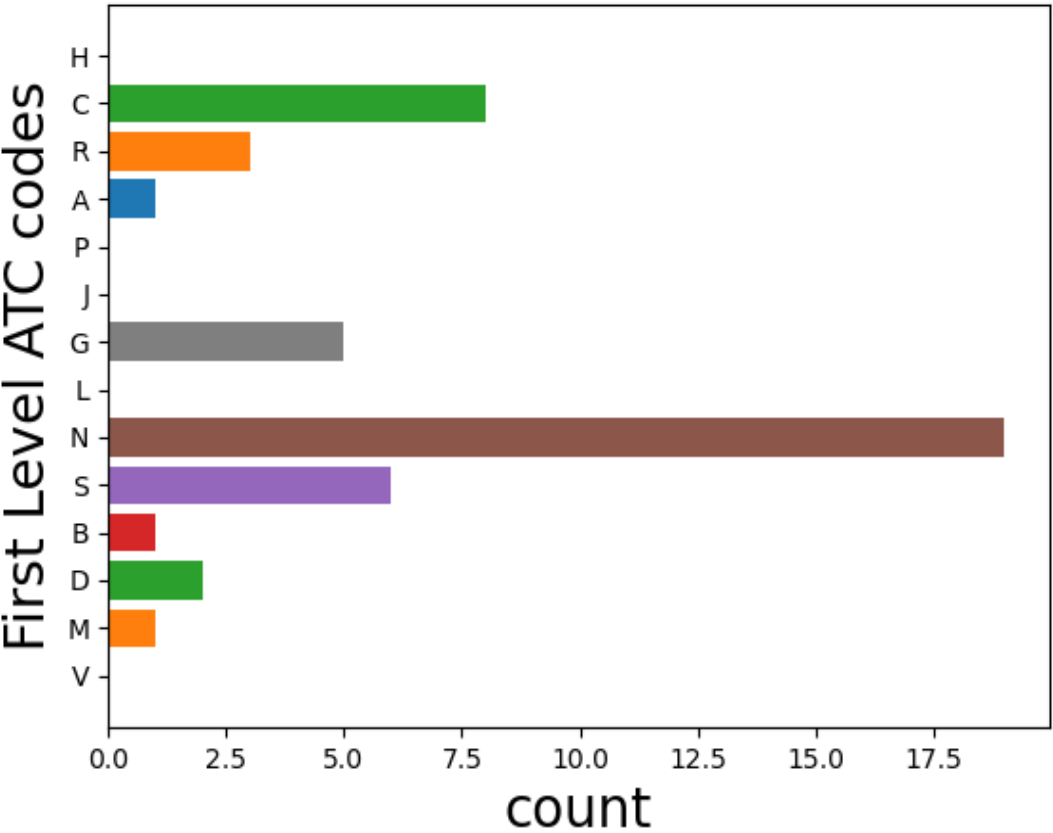
D. Colibășanu, V. Groza, M.A. Occhiuzzi, F. Grande, M. Udrescu, L. Udrescu



Community/Cluster 9:

{'V': set(), 'M': {'Tolazoline'}, 'D': {'Oxymetazoline', 'Brimonidine'}, 'B': {'Epinephrine'}, 'S': {'Brimonidine', 'Epinephrine', 'Apraclonidine', 'Xylometazoline', 'Clonidine', 'Oxymetazoline'}, 'N': {'Cabergoline', 'Ziprasidone', 'Rotigotine', 'Trimipramine', 'Lisuride', 'Clozapine', 'Pizotifen', 'Methotrimeprazine', 'Quetiapine', 'Metamfetamine', 'Apomorphine', 'Asenapine', 'Clonidine', 'Aripiprazole', 'Paliperidone', 'Mianserin', 'Loxapine', 'Bromocriptine', 'Risperidone'}, 'L': set(), 'G': {'Cabergoline', 'Lisuride', 'Apomorphine', 'Yohimbine', 'Bromocriptine'}, 'J': set(), 'P': set(), 'A': {'Epinephrine'}, 'R': {'Oxymetazoline', 'Epinephrine', 'Xylometazoline'}, 'C': {'Droxidopa', 'Norepinephrine', 'Fenoldopam', 'Guanfacine', 'Epinephrine', 'Tolazoline', 'Celiprolol', 'Clonidine'}, 'H': set()}

In [3]:



Drugs with property N:

['Cabergoline', 'Ziprasidone', 'Rotigotine', 'Trimipramine', 'Lisuride', 'Clozapine', 'Pizotifen', 'Methotrimeprazine', 'Quetiapine', 'Metamfetamine', 'Apomorphine', 'Asenapine', 'Clonidine', 'Aripiprazole', 'Paliperidone', 'Mianserin', 'Loxapine', 'Bromocriptine', 'Risperidone']

In [5]:

Drugs not having property N:

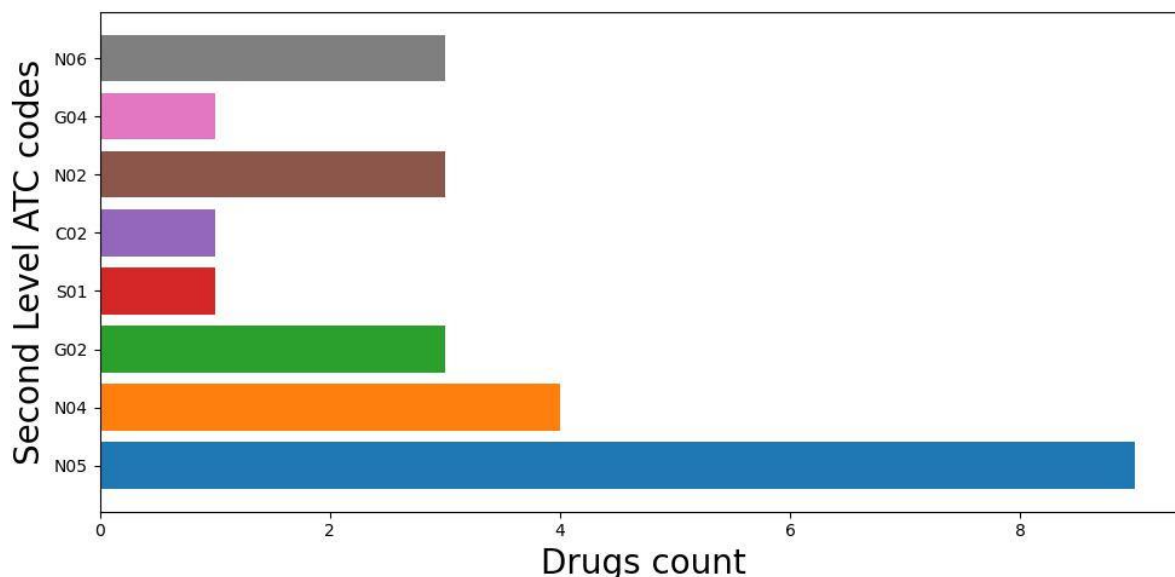
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Accuracy: 61.29032258064516

In [6]:

Level 2 on N:

{'N05': {'Quetiapine', 'Ziprasidone', 'Paliperidone', 'Asenapine', 'Clozapine', 'Loxapine', 'Aripiprazole', 'Risperidone', 'Methotrimeprazine'}, 'N04': {'Cabergoline', 'Bromocriptine', 'Apomorphine', 'Rotigotine'}, 'G02': {'Cabergoline', 'Bromocriptine', 'Lisuride'}, 'S01': {'Clonidine'}, 'C02': {'Clonidine'}, 'N02': {'Lisuride', 'Pizotifen', 'Clonidine'}, 'G04': {'Apomorphine'}, 'N06': {'Metamfetamine', 'Trimipramine', 'Mianserin'}}

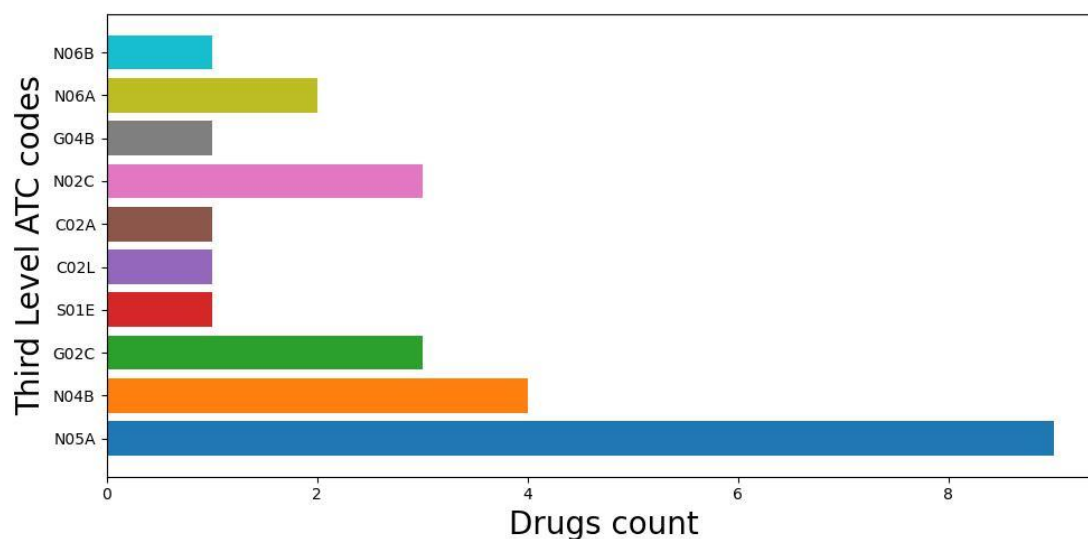


Nivelul 3 pe N:

{'N05A': {'Quetiapine', 'Ziprasidone', 'Paliperidone', 'Asenapine', 'Clozapine', 'Loxapine', 'Aripiprazole', 'Risperidone', 'Methotrimeprazine'}, 'N04B': {'Cabergoline', 'Bromocriptine', 'Apomorphine', 'Rotigotine'}, 'G02C': {'Cabergoline', 'Bromocriptine', 'Lisuride'}, 'S01E': {'Clonidine'}, 'C02L': {'Clonidine'}, 'C02A': {'Clonidine'}, 'N02C': {'Lisuride', 'Pizotifen', 'Clonidine'}, 'G04B': {'Apomorphine'}, 'N06A': {'Mianserin', 'Trimipramine'}, 'N06B': {'Metamfetamine'}}

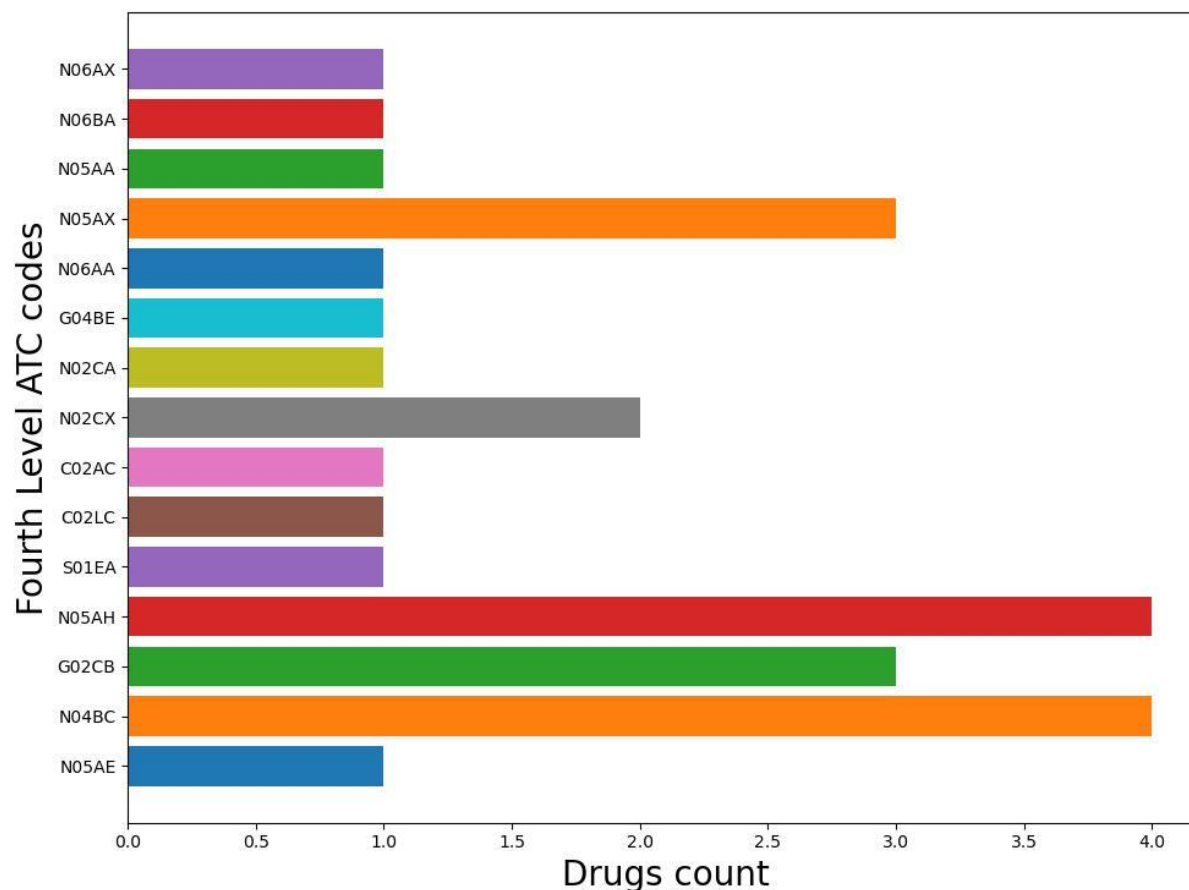
Drug Repositioning Pipeline Integrating Community Analysis in Drug-Drug Similarity Networks and Automated ATC Community Labeling to Enhance Molecular Docking Validation

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Level 4 on N:

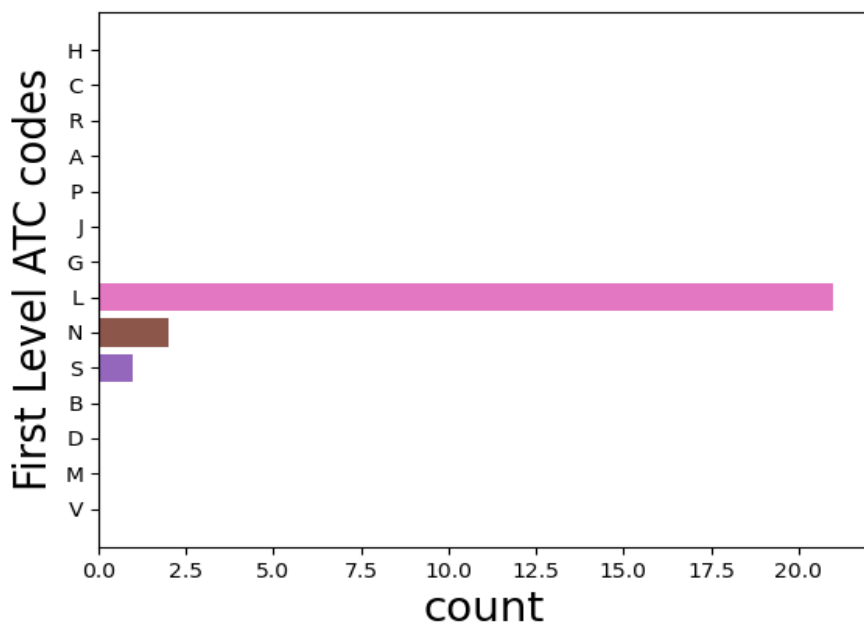
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Community/Cluster 10:

```
{'V': set(), 'M': set(), 'D': set(), 'B': set(), 'S': {'Cenegermin'}, 'N': {'Amitriptyline', 'Rufinamide'}, 'L': {'Gefitinib', 'Erlotinib', 'Larotrectinib', 'Panitumumab', 'Gilteritinib', 'Afatinib', 'Neratinib', 'Lapatinib', 'Cabozantinib', 'Dacomitinib', 'Capmatinib', 'Alectinib', 'Amivantamab', 'Necitumumab', 'Ceritinib', 'Tepotinib', 'Osimertinib', 'Vandetanib', 'Lorlatinib', 'Crizotinib', 'Entrectinib'}, 'G': set(), 'J': set(), 'P': set(), 'A': set(), 'R': set(), 'C': set(), 'H': set()}
```

In [3]:



Drugs with property L:

```
['Gefitinib', 'Erlotinib', 'Larotrectinib', 'Panitumumab', 'Gilteritinib', 'Afatinib', 'Neratinib', 'Lapatinib', 'Cabozantinib', 'Dacomitinib', 'Capmatinib', 'Alectinib', 'Amivantamab', 'Necitumumab', 'Ceritinib', 'Tepotinib', 'Osimertinib', 'Vandetanib', 'Lorlatinib', 'Crizotinib', 'Entrectinib']
```

In [5]:

Drugs not having property L:

```
['Amitriptyline', 'Rufinamide', 'Cenegermin', 'Amitriptyline', 'Rufinamide', 'Cenegermin']
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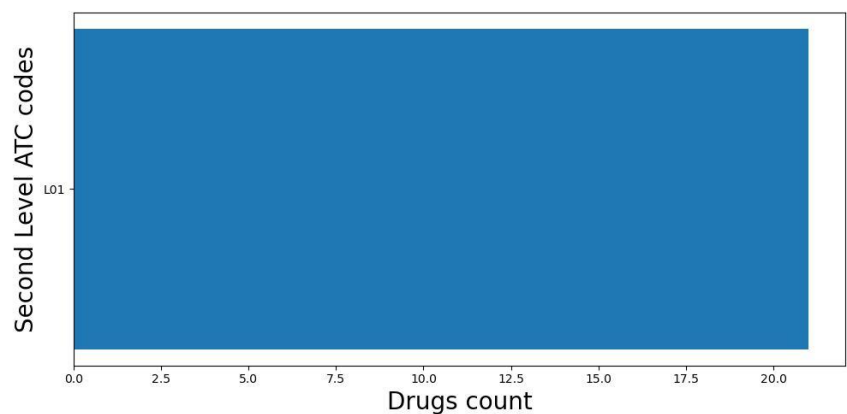
Accuracy: 87.5

Level 2 on L:

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{'L01': {'Gefitinib', 'Erlotinib', 'Larotrectinib', 'Panitumumab', 'Gilteritinib', 'Afatinib', 'Neratinib', 'Lapatinib', 'Cabozantinib', 'Dacomitinib', 'Capmatinib', 'Alectinib', 'Amivantamab', 'Necitumumab', 'Ceritinib', 'Tepotinib', 'Osimertinib', 'Vandetanib', 'Lorlatinib', 'Crizotinib', 'Entrectinib'}}
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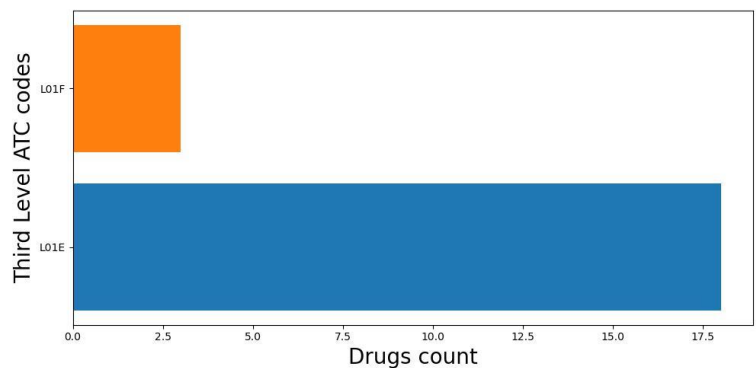

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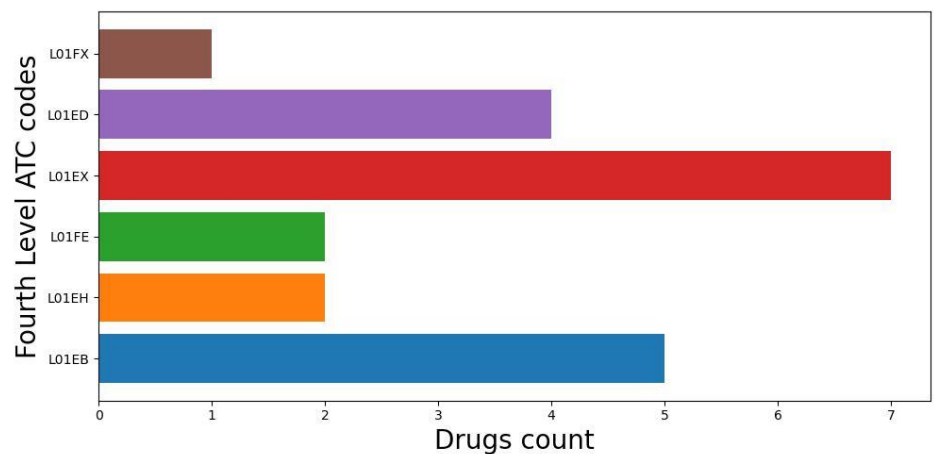
Level 3 on L:

{'L01E': {'Alectinib', 'Gilteritinib', 'Gefitinib', 'Ceritinib', 'Tepotinib', 'Neratinib', 'Erlotinib', 'Vandetanib', 'Lapatinib', 'Lorlatinib', 'Crizotinib', 'Larotrectinib', 'Cabozantinib', 'Dacomitinib', 'Entrectinib', 'Osimertinib', 'Afatinib', 'Capmatinib'}, 'L01F': {'Panitumumab', 'Necitumumab', 'Amivantamab'}}



Level 4 on L:

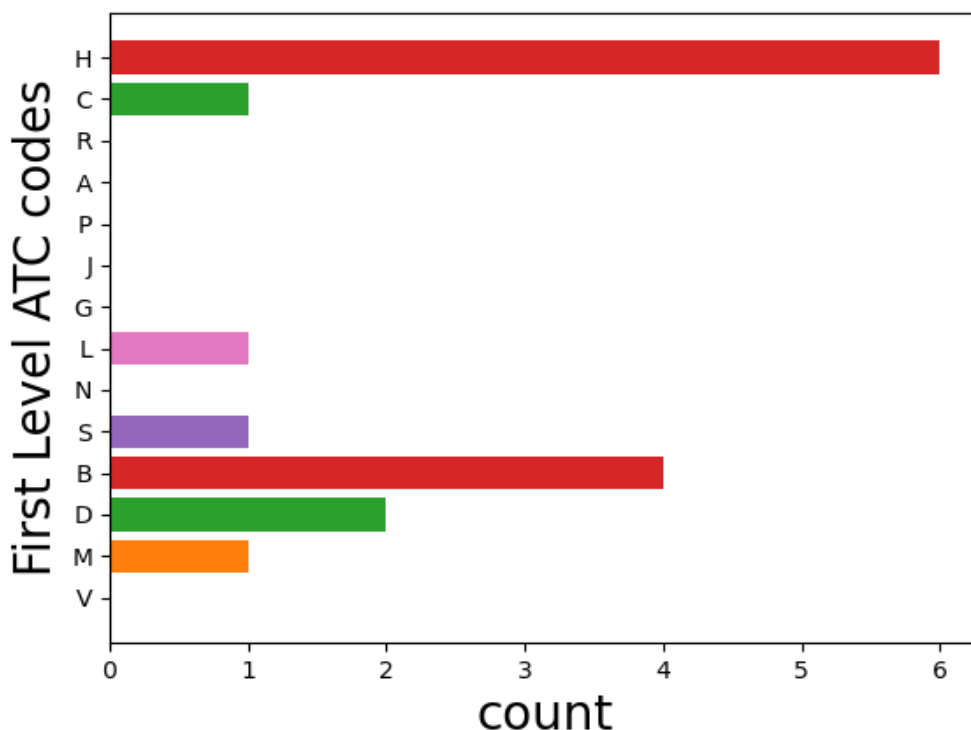
{'L01EB': {'Gefitinib', 'Erlotinib', 'Dacomitinib', 'Osimertinib', 'Afatinib'}, 'L01EH': {'Neratinib', 'Lapatinib'}, 'L01FE': {'Panitumumab', 'Necitumumab'}, 'L01EX': {'Tepotinib', 'Vandetanib', 'Larotrectinib', 'Cabozantinib', 'Entrectinib', 'Gilteritinib', 'Capmatinib'}, 'L01ED': {'Alectinib', 'Crizotinib', 'Lorlatinib', 'Ceritinib'}, 'L01FX': {'Amivantamab'}}



Community/Cluster 11:

{'V': set(), 'M': {'Cisatracurium'}, 'D': {'Resorcinol', 'Triclosan'}, 'B': {'Eptifibatide', 'Abciximab', 'Ferric maltol', 'Tirofiban'}, 'S': {'Resorcinol'}, 'N': set(), 'L': {'Antithymocyte immunoglobulin (rabbit)'}, 'G': set(), 'J': set(), 'P': set(), 'A': set(), 'R': set(), 'C': {'Dextrothyroxine'}, 'H': {'Thyrotropin alfa', 'Propylthiouracil', 'Liothyronine', 'Levothyroxine', 'Carbimazole', 'Methimazole'}}

In [3]:



Drugs with property H:

['Thyrotropin alfa', 'Propylthiouracil', 'Liothyronine', 'Levothyroxine', 'Carbimazole', 'Methimazole']

Drugs not having property H:

['Abciximab', 'Eptifibatide', 'Antithymocyte immunoglobulin (rabbit)', 'Dextrothyroxine', 'Cisatracurium', 'Tirofiban', 'Triclosan', 'Resorcinol', 'Ferric maltol', 'Eptifibatide', 'Abciximab', 'Antithymocyte immunoglobulin (rabbit)', 'Dextrothyroxine', 'Cisatracurium', 'Tirofiban', 'Triclosan', 'Resorcinol', 'Ferric maltol']

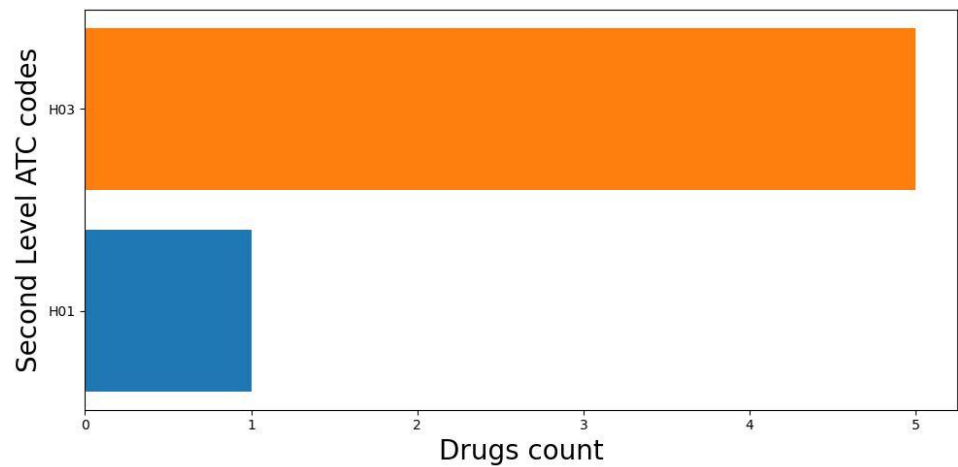
Accuracy: 40

Level 2 on H:

{'H01': {'Thyrotropin alfa'}, 'H03': {'Propylthiouracil', 'Liothyronine', 'Levothyroxine', 'Carbimazole', 'Methimazole'}}

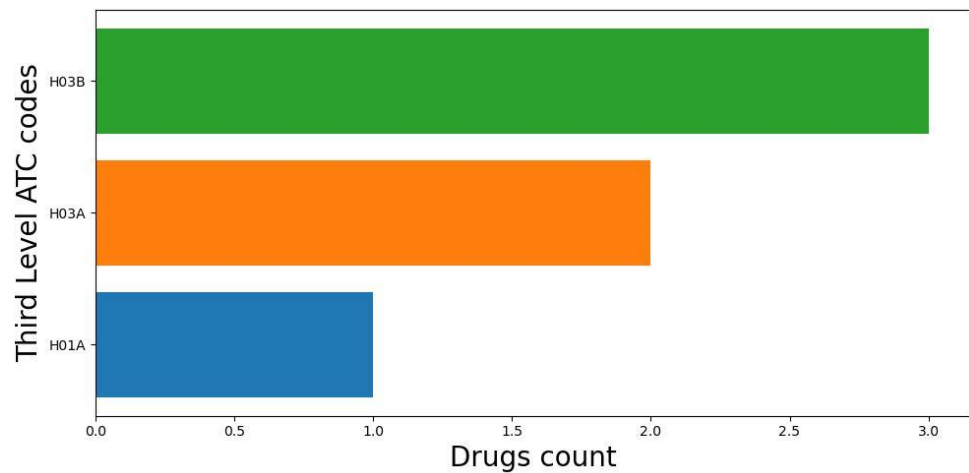
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D. Colibășanu, V. Groza, M.A. Occhiuzzi, F. Grande, M. Udrescu, L. Udrescu



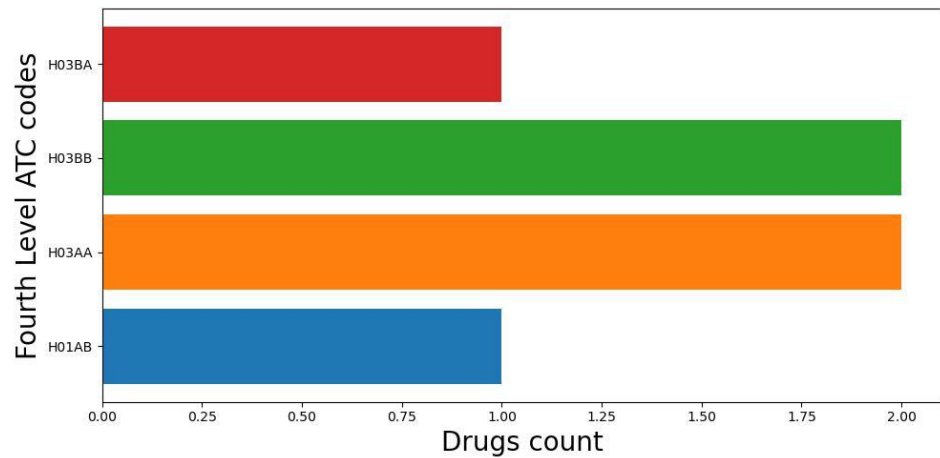
Level 3 on H:

{'H01A': {'Thyrotropin alfa'}, 'H03A': {'Levothyroxine', 'Liothyronine'}, 'H03B': {'Methimazole', 'Carbimazole', 'Propylthiouracil'}}



Level 4 on H:

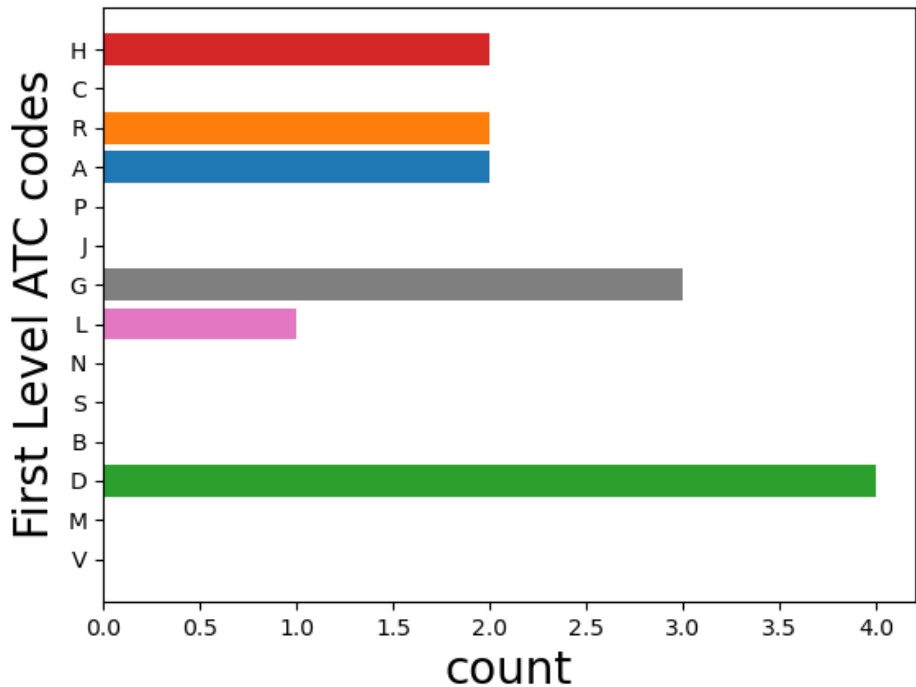
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Community/Cluster 12:

{'V': set(), 'M': set(), 'D': {'Finasteride', 'Hydroquinone', 'Monobenzzone', 'Azelaic acid'}, 'B': set(), 'S': set(), 'N': set(), 'L': {'Abiraterone'}, 'G': {'Finasteride', 'Dutasteride', 'Bremelanotide'}, 'J': set(), 'P': set(), 'A': {'Metformin', 'Glycyrrhizic acid'}, 'R': {'Hexylresorcinol', 'Tyloxapol'}, 'C': set(), 'H': {'Corticotropin', 'Tetracosactide'}}

In [3]:



Drugs with property D:

['Finasteride', 'Hydroquinone', 'Monobenzzone', 'Azelaic acid']

Drugs not having property D:

['Metformin', 'Dutasteride', 'Tetracosactide', 'Corticotropin', 'Abiraterone', 'Tyloxapol', 'Hexylresorcinol', 'Bremelanotide', 'Glycyrrhizic acid', 'Metformin', 'Dutasteride', 'Tetracosactide', 'Corticotropin', 'Abiraterone', 'Tyloxapol', 'Hexylresorcinol', 'Bremelanotide', 'Glycyrrhizic acid']

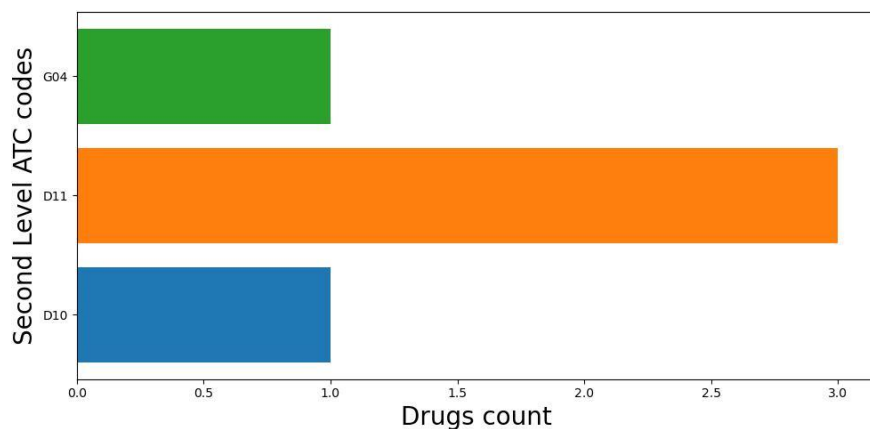
Accuracy: 30.76923076923077

Level 2 on D:

{'D10': {'Azelaic acid'}, 'D11': {'Finasteride', 'Hydroquinone', 'Monobenzzone'}, 'G04': {'Finasteride'}}

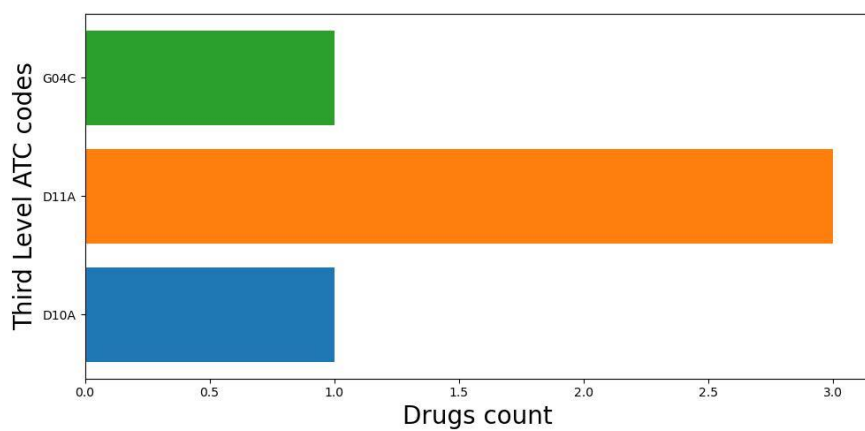
Drug Repositioning Pipeline Integrating Community Analysis in Drug-Drug Similarity Networks and Automated ATC Community Labeling to Enhance Molecular Docking Validation

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Level 3 on D:

{'D10A': {'Azelaic acid'}, 'D11A': {'Finasteride', 'Hydroquinone', 'Monobenzone'}, 'G04C': {'Finasteride'}}



Level 4 on D:

{'D10AX': {'Azelaic acid'}, 'D11AX': {'Finasteride', 'Hydroquinone', 'Monobenzone'}, 'G04CA': {'Finasteride'}, 'G04CB': {'Finasteride'}}

